

Cloud Platform Reference Designs

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Cloud Platform Reference Designs

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Date	Revision	Author	Description
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2019.11.05	01.00.03	Guy Turgeon (Wind River)	Initial Cloud Platform Requirements
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2020.03.04	Baseline-v01.00.07	Arjun Nanjundappa (Mavenir)	Included both In-line and Look-aside acceleration diagrams for O-DU and Accelerators to other nodes have been kept generic for section 4.3.2.
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Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN ALLIANCE.

The contents of the present document are subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of release date and an increase in version number as follows:

Release x.y.z

where:

x the first digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).

y the second digit is incremented when editorial only changes have been incorporated in the document.

z the third digit included only in working versions of the document indicating incremental changes during the editing process.

The present document focuses on cloud deployment scenario B and C (including C.1 and C.2), as defined in the O-RAN WG6 Cloud Architecture document [1], and specifies requirements and reference designs for the cloud platform hardware and software.

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of this specification.

- [1] O-RAN WG6, Cloud Architecture and Deployment Scenarios for O-RAN Virtualized RAN.
- [2] O-RAN WG4, Control, User and Synchronization Plane Specification.
- [3] GR-63-CORE for Network Switching Systems (NEBS) Seismic Rated Enclosures, Telcordia.
- [4] GR-1089-CORE Electromagnetic Compatibility and Electrical Safety, Telcordia.
- [5] OTII Server Technical Specification, 2019.
- [6] OCP openEDGE Project, <https://www.opencompute.org/wiki/Telcos/openEDGE>.
- [7] ETSI EN300 019-1-3 Class 3.2 Equipment Engineering (EE) Environmental Conditions and Environmental Tests for Telecommunications Equipment Part 1-3: Classification of Environmental Conditions Stationary Use at Weather-protected Locations.
- [8] ETSI EN300386 (v1.6.1) Terrestrial Trunked Radio (TETRA); Direct Mode Operation (DMO); Part 6: Security
- [9] FCC CFR47 15 (class A), FCC, Rules and Regulations for Title 47 Telecommunications.
- [10] CISPR 22/32 (Class A) International Electrotechnical Commission (IEC) International Special Committee on Radio Interference, Radiated Emission Limits.
- [11] CISPR 24, IEC Information Technology Equipment – Immunity Characteristics.
- [12] TEC/EMI/TEL-001/01/FEB-09 – Information Technology Equipment.
- [13] TEC/IR/SWN-2MB/07/MAR-10 – Information Technology Equipment.
- [14] IEC 62368-1:2014, Safety Standard for Information Technology Equipment and Audio Video Equipment.
- [15] O-RAN WG4 Management Plane Specification.
- [16] 3GPP TS 23.501: “System architecture for the 5G System (5GS)”.
- [17] O-RAN WG8, Base Station O-DU and O-CU Software Architecture and APIs.
- [18] O-RAN Infrastructure Project, <https://wiki.o-ran-sc.org/display/IN/Infrastructure+Home>.

1.3 Abbreviations

For the purposes of the present document, the following abbreviations apply.

AAL	Acceleration Abstraction Layer
API	Application Programming Interface
ASIC	Application-specific Integrated Circuit
BIOS	Basic Input Output System
BBU	Base Band Unit
CNF	Containerized Network Function
CPU	Central Processing Unit
DSP	Digital Signal Processing
FPGA	Field Programmable Gate Array
DPDK	Data Plane Development Kit
eNB	eNodeB (applies to LTE)
EPA	Enhanced Platform Awareness
FEC	Forward Error Correcting
FFS	For Further Study
FFT	Fast Fourier Transform
FHGW	Front-haul Gateway
FTS	Full Timing Support
gNB	gNodeB (applies to NR)
GNSS	Global Navigation Satellite System
GPU	Graphics Processing Unit
HA	High Availability
HW	Hardware
LLS-C3	Lower Layer Split Configuration 3
NFV	Network Functions Virtualization
NFVI	Network Functions Virtualization Infrastructure
NIC	Network Interface Card
NSA	Non-standalone
NUMA	Non-uniform Memory Access
O-CU	O-RAN Centralized Unit
O-DU	O-RAN Distributed Unit
O-RU	O-RAN Radio Unit
OS	Operating System
PHC	PTP Hardware Clock
PIM	Physical Infrastructure Manager
PRACH	Physical Random Access Channel
PRTC	Primary Reference Time Clock
PSU	Power Supply Unit
PTP	Precision Time Protocol

194	PTS	Partial Timing Support
195	RAID	Redundant Array of Inexpensive Disks
196	RIC	RAN Intelligent Controller
197	RT	Real-time
198	SA	Standalone
199	SMO	Service Management and Orchestration
200	SR-IOV	Single Root I/O Virtualization
201	SW	Software
202	T-BC	Telecom Boundary Clock
203	T-GM	Telecom Grand Master
204	T-TSC	Telecom Time Slave Clock
205	UEFI	Unified Extensible Firmware Interface
206	vBBU	Virtual Baseband Unit
207	VF	Virtual Function
208	VIM	Virtualized Infrastructure Manager
209	VM	Virtual Machine
210	VNF	Virtualized Network Function
211		

Chapter 2 Deployment Scenarios

This document focuses on cloud deployment scenarios B and C (including both variants C.1 and C.2), as defined in the O-RAN Cloud Architecture document [1]. The current version of this document focuses on SA NR gNBs deployed on the O-Cloud. The extensions of O-Cloud, if required, to support NSA scenarios indicated in the annexes of the O-RAN Cloud Architecture document [1] are FFS.

O-Clouds are deployed at both regional and edge locations. At a given location, the edge O-Cloud can scale from a single server to a clustered configuration with possibly hundreds of servers. This version of the document focuses on a container-based cloud platform (see Ch.5.2 for a reference design example) and it will address VM-based designs in a future release. For the time synchronization aspects, this version of the document focuses on the LLS-C3 configuration [2], i.e., O-DU and O-RU are synchronized from the fronthaul network (see Ch.5.3 for a reference design example).

The terms, vO-CU and vO-DU, represent virtualized or containerized O-CU and O-DU, and are used interchangeably with O-CU and O-DU in this document. The optional use of a Front-haul Gateway (FHGW) in the architecture allows significant savings in providing transport between the O-RU and O-DU functionalities.

2.1 Scenario B

Figure 2-1 illustrates scenario B, where the near-RT RIC is virtualized or containerized on a regional O-Cloud, and the O-CU and O-DU functions are virtualized or containerized on an edge O-Cloud that typically resides at a different location.

The interface that the near-RT RIC uses between the regional O-Cloud and the edge O-Cloud is E2. Interfaces between the O-CU and O-DU managed functions are either within the same single managed element, or among multiple managed elements within the same edge O-Cloud.

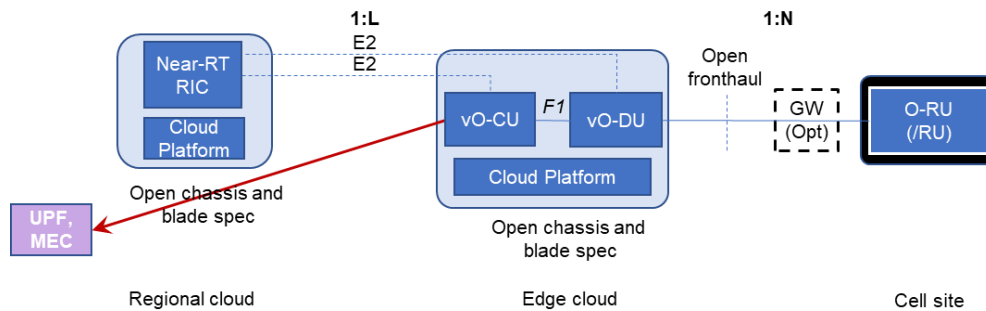


Figure 2-1. Cloud deployment scenario B.

This scenario is to support deployments in locations with limited remote front-haul capacity and a set of O-RUs spread out in a limited area, supported by pooled vO-CU/vO-DU functionality while still meeting the O-DU latency requirements. The optional use of a Front-haul Gateway (FHGW) in the architecture allows significant savings in providing transport between the O-RU and vO-DU functionalities.

Figure 2-2 shows the O-Cloud high-level deployment diagram. One vO-CU and one vO-DU together form a virtual BBU (vBBU), which connects to N O-RUs (typically up to 64, see [1]). This deployment should support tens of thousands of O-RUs per near-RT RIC, and therefore L could easily exceed 100.

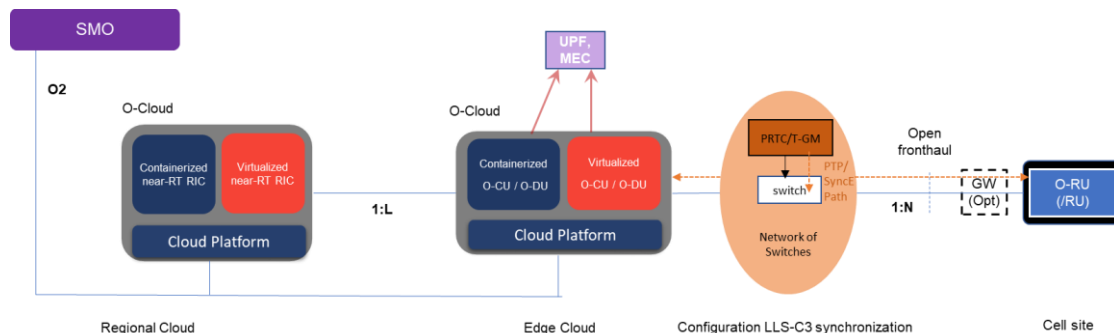


Figure 2-2. Scenario B – O-Cloud high-level deployment diagram.

2.2 Scenario C

Figure 2-3 illustrates scenario C, where the near-RT RIC and the vO-CU functions are virtualized or containerized on a regional O-Cloud, and the O-DU function is virtualized or containerized on an edge O-Cloud that typically resides at a different location.

The interfaces that the near-RT RIC and O-CU use between the regional O-Cloud and the edge O-Cloud are E2 and, respectively, F1 (i.e., F1-c and F1-u). Interfaces between the near-RT RIC and O-CU managed functions are either within the same single managed element, or among multiple managed elements within the same regional O-Cloud.

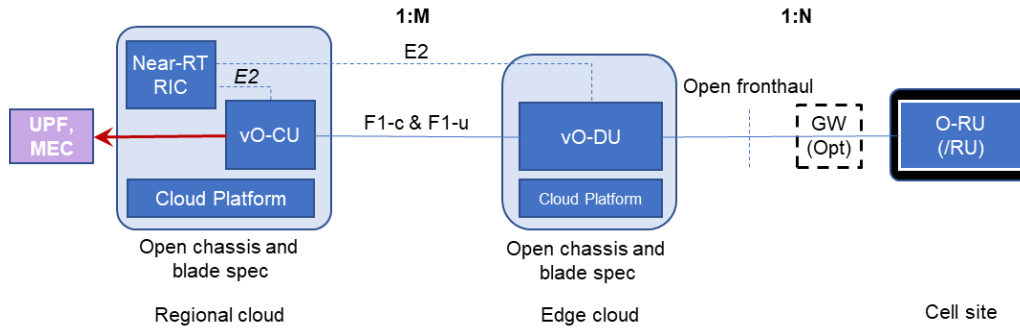


Figure 2-3. Cloud deployment scenario C.

This scenario is to support deployments in locations with limited remote front-haul capacity and a set of O-RUs spread out in a small area that restricts the number of O-RUs that can be pooled while still meeting the O-DU latency requirements. The O-CU function is further pooled to increase the efficiency of the hardware platform which it shares with the near-RT RIC function.

Figure 2-4 shows the O-Cloud high-level deployment diagram. The edge O-Cloud is expected to connect to N O-RUs (typically up to 64, see [1]). This deployment should support tens of thousands of O-RUs per near-RT RIC, and therefore M could easily exceed 100.

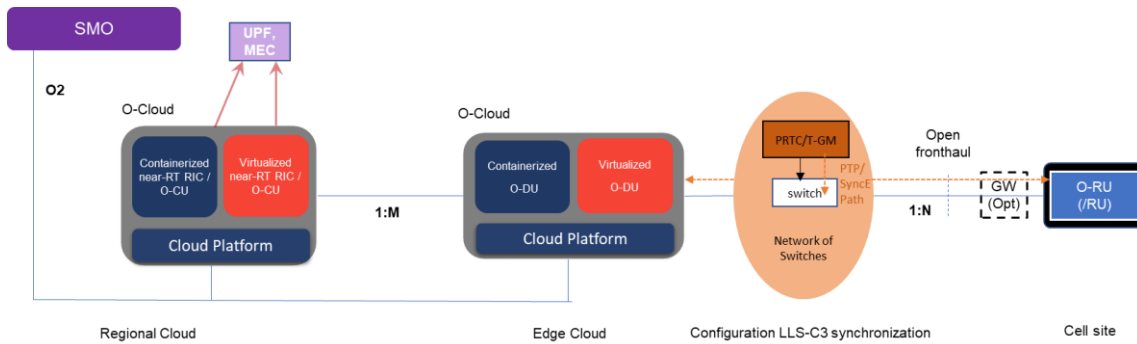


Figure 2-4. Scenario C – O-Cloud high-level deployment diagram.

2.2.1 Scenario C.1

This is a variation of Scenario C, driven by the fact that different types of traffic (network slices) have different latency requirements. In particular, URLLC has more demanding user-plane latency requirements, and Figure 2-5 below shows how the vO-CU User Plane (vO-CU-UP) could be terminated in different places for different network slices. Below, network slice 3 is terminated in the edge O-Cloud. This scenario is also suitable in case there is not enough space or power supply to install all vO-CUs and vO-DUs in one edge O-Cloud.

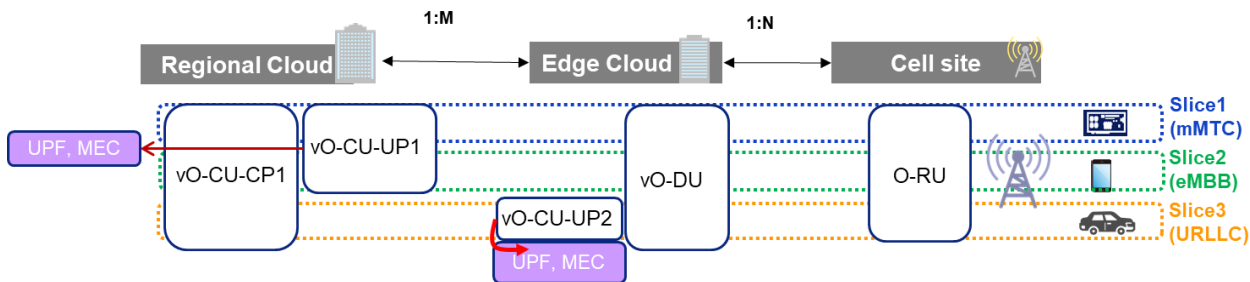


Figure 2-5. Treatment of Network Slice: MEC for URLLC at Edge Cloud, Centralized Control, Single vO-DUs

In Scenario C.1, all O-CU control is placed in the regional O-Cloud, and there is a single vO-DU for all network slices. Only the placement of the vO-CU-UP differs, depending on the network slice. The diagram of how network functions map to network elements for scenario C.1 is shown in Figure 2-6 (please note that in Scenario C.1, the E1 interface between CU-CP and CU-UP is explicitly used, unlike in Scenario C or C.2).

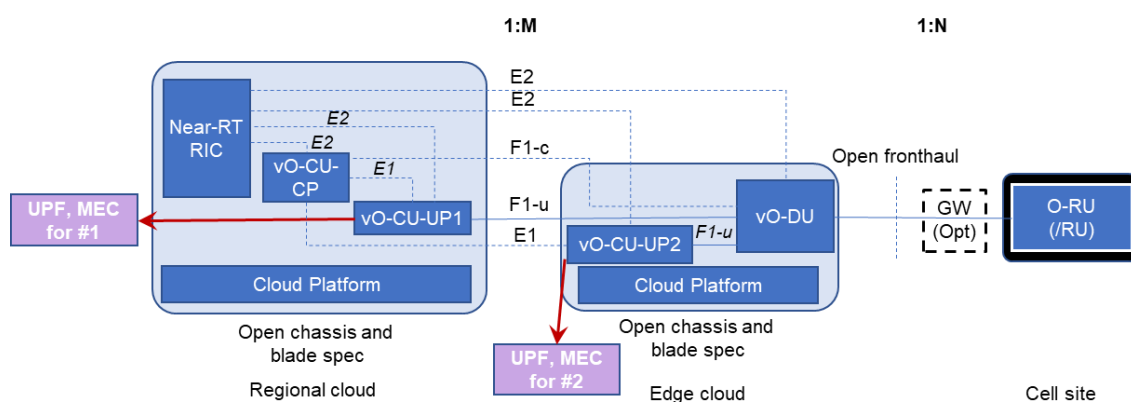


Figure 2-6. Cloud deployment scenario C.1.

Table 2-1 is a summary of the cardinality and the distance/delay requirements assumed for this scenario. The URLLC user plane requirements are what drive the placement of the vO-CU-UP function to be in the edge O-Cloud.

Table 2-1. Cardinality and Delay Performance for Scenario C.1

	Attribute	RIC – O-CU	O-CU – O-DU	O-DU – O-RU/RU
	Example Cardinality	L= 1	M=320	N=100
Delay Max 1-way (distance)	mMTC	NA	625 μs (125 km)	100 μs (20 km)
	eMBB	NA	625 μs (125 km)	100 μs (20 km)
	URLLC (user/control)	NA	100 μs (20 km)/625 μs (125 km)	100 μs (20 km)

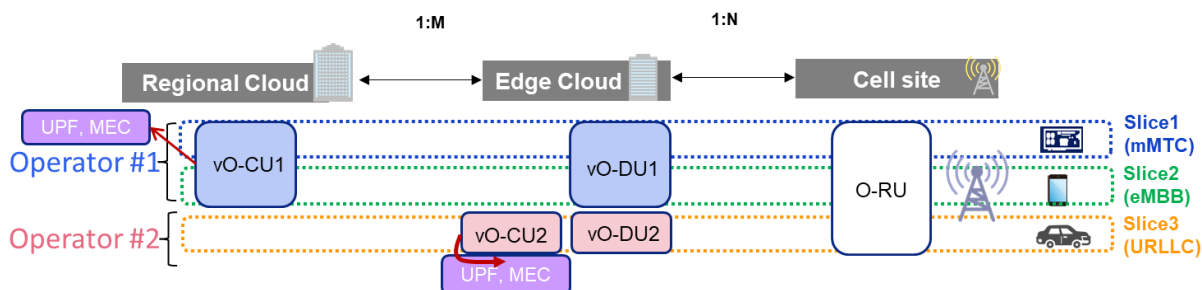
2.2.2 Scenario C.2

This is a second variation of Scenario C, which utilizes the same method of placing some vO-CU user plane functionality in the edge O-Cloud, and some in the regional O-Cloud. However, instead of having one vO-DU for all network slices, there are different vO-DU instances in the edge O-Cloud.

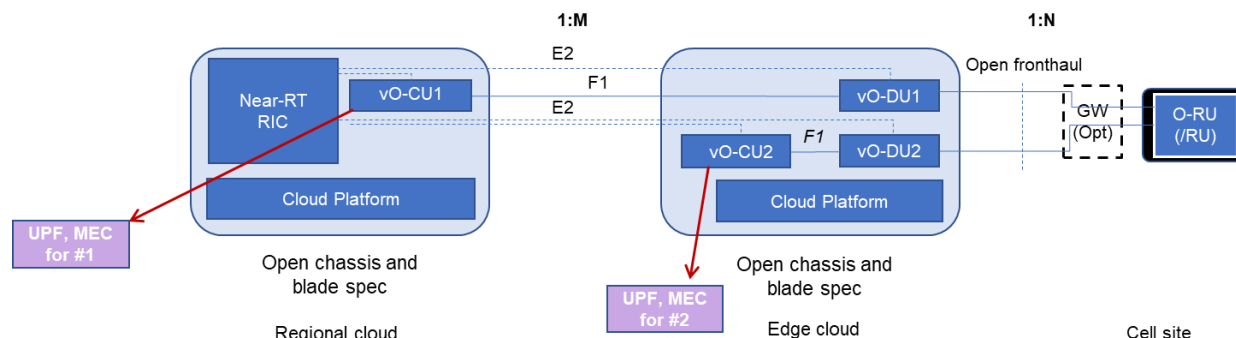
It is driven by factors including the following two use cases:

- One driver is RAN (O-RU) sharing among operators. In this use case, any operator can flexibly launch vO-CU and vO-DU instances at edge or regional O-Cloud site. For example, as shown in Figure 2-7, Operator #1 wants to launch the vO-CU1 instance in the regional O-Cloud, and the vO-DU1 instance at subtending edge O-Cloud

- Another driver is that, even within a single operator, that operator can customize scheduler functions depending on the network slice types, and can place the vO-CU and vO-DU instances depending on the network slice types. For example, an operator may launch both vO-CU and vO-DU at the edge O-Cloud site (see Operator #2 below) to provide a URLLC service.



The diagram of how network functions map to network elements for scenario C.2 is shown in Figure 2-8.



The performance requirements are the same as those discussed earlier for Scenario C.1 in Section 2.2.1, with one exception, namely, for the URLLC slice in Scenario C.2 the CU-CP and CU-UP delays are the same (i.e., they are collocated on the edge cloud in this particular example).

Carrier Type	Component Carrier Bandwidth	Number of Layers (Downlink)	Number of Aggregated Carriers	Total Aggregated Bandwidth
LTE	20 MHz	4 layers	5 carriers	400 MHz
NR Sub-6 GHz	20 MHz	16 layers	5 carriers	1600 MHz
mmWave	100 MHz	2 layers	8 carriers	1600 MHz

Carrier Type	Component Carrier Bandwidth	Number of Layers	Number of Aggregated Carriers	Total Aggregated Bandwidth
LTE	5-20 MHz	2, 4 layers	3-5 carriers	80-400 MHz
NR Sub-6 GHz	20-100 MHz	4 - 16 layers	1-3 carriers	Up to 2240 MHz
mmWave	FFS	FFS	FFS	FFS

Table 3-2. Deutsche Telekom carrier aggregation requirements.

Carrier Type	Component Carrier Bandwidth	Number of Layers	Number of Aggregated Carriers	Total Aggregated Bandwidth
LTE	20 MHz	4 layers	4 carriers	320 MHz
NR Sub-6 GHz	100 MHz	8 or 16 layers	1 carrier	Up to 1600 MHz
mmWave	FFS	FFS	FFS	FFS

Table 3-3. Orange carrier aggregation requirements.

Carrier Type	Component Carrier Bandwidth	Number of Layers (Downlink)	Number of Aggregated Carriers	Total Aggregated Bandwidth
LTE	5-20 MHz	4 layers	3-5 carriers	Up to 400 MHz
NR Sub-6 GHz	100 MHz	8 – 16 layers	2 carriers	Up to 3200 MHz
mmWave	100 MHz	2 layers	4 carriers	800 MHz

Table 3-4. KDDI carrier aggregation requirements.

Carrier Type	Component Carrier Bandwidth	Number of Layers (Downlink)	Number of Aggregated Carriers	Total Aggregated Bandwidth
LTE	10MHz, 20MHz	2, 4	2, 3, 4 carriers	30~80MHz
NR Sub-6 GHz	30MHz, 60MHz, 100MHz	2, 4	2 carriers	130~200MHz
mmWave	FFS	FFS	FFS	FFS

Table 3-5. CMCC carrier aggregation requirements.

Carrier Type	Component Carrier Bandwidth	Number of Layers (Downlink)	Number of Aggregated Carriers	Total Aggregated Bandwidth
LTE	5MHz, 10MHz, 15MHz, 20MHz	2	2, 3 carriers	Up to 50MHz
NR Sub-6 GHz	40MHz, 50MHz, 55MHz, 100MHz	2, 4	2 , 3 carriers	Up to 300MHz
mmWave	FFS	FFS	FFS	FFS

Table 3-6. CTC carrier aggregation requirements.

Active UEs (95 th percentile) supported per macro sub-6 GHz TRP	256 (AT&T, others FFS)
Active UEs (95th percentile) supported per mmWave TRP	64 (AT&T, others FFS)
Carrier capacity (MHz) per O-DU pool	Up to 56,000 MHz (AT&T, others FFS)
O-DU pooling model	Pooled centralization [1]
Minimum number of TRPs that an O-DU pool supports	32 (AT&T and Orange, others FFS)
Fronthaul interface to O-DU	O-RAN 7.2x, with low phy in O-RU or FHGW
Physical interface to the cloud platform supporting O-DU (e.g., from FHGW)	At least 2x 25Gbps (or higher depending on site configuration) aggregated links to minimize switching latency (2x is for redundancy purposes).

Table 3-7. BBU pooling requirements.

3.2 Cloud Deployment Requirements

As Figure 2-1 shows, deployment of a cloud platform occurs in two locations: regional and edge clouds. The physical environment of a regional cloud is typically a modern data center, hosting standard server racks with a 19-inch wide rack enclosure, 42U height, and external depth between 1 and 1.2 meters. In this environment, the cloud platform hardware can use standard rack-mount servers and switches, and does not impose special requirements on the hardware form factors.

The edge cloud often resides in a traditional telecommunications equipment room with limited space, cooling capacity, floor loading, and so forth. The limitations can impose special requirements on the cloud platform hardware that may require telecommunications standard certifications, such as the Network Equipment Building System (NEBS) requirements found in GR-63-CORE [3] and GR-1089-CORE [4]. For example, the Open Telecom IT Infrastructure (OTII) Project specifies a 2U rack server with depth less than 450 mm [5][3]. In addition, the Open Compute Project (OCP) operates an openEDGE project [6], which specifies a compact cloud platform that has the extended environmental requirements often needed in edge deployments.

3.2.1 Regional Cloud

A regional cloud hosts the near-RT RIC, which includes Radio Connection Management, Mobility Management, QoS Management, Interference Management, Radio-Network Information Base, and possibly 3rd-party applications. A typical regional cloud has the following characteristics:

- Standard Internet Data Center (IDC) environment;
- At least one standard 19-inch rack with a top-of-rack (TOR) switch and five servers;
- Highly available cloud platform software;
- Highly available network topology;
- Highly available storage system in the form of traditional storage area network (SAN), software-defined storage (SDS), or combination of the two.

In Scenario C (and its variants C.1/C.2), a regional cloud will also host the O-CU functions.

3.2.2 Edge Cloud

The edge cloud is often deployed in a relatively rough physical environment with typical characteristics in Table 3-8.

Rack Form Factors	19-inch width, 600mm depth
Temperature Range	Long-term: -5°C to 45°C (ETSI EN300 019-1-3 Class 3.2 [7]) Short-term: -5°C to 55°C (GR-63-CORE [3])
Humidity	Operating humidity: 5% to 95% Long-term: 5% to 85% Short-term: 5% to 90%

EMC	EN300386 (v1.6.1) [8] FCC CFR47 15 (class A) [9], CISPR 22/32 (class A) [10], CISPR 24 [11] TEC/EMI/TEL-001/01/FEB-09 [12] and TEC/IR/SWN-2MB/07/MAR-10 [13] GR-1089-CORE [4]
Equipment Center	Class A (Telecom center); Class B (Non-telecom center)
Seismic	GR-63-CORE (Zone 4) [3]
Safety	IEC 62368-1:2014 [14] GR-63-CORE (Electrical safety, grounding and bonding) [3]
Acoustic Noise	GR-63-CORE (equipment room criteria) [3]
Fire Resistance	GR-63-CORE (shelf level criteria) [3]
Power Supply	208V AC, 220V AC, or -48V DC (-60V DC in Germany)

Table 3-8. Edge cloud environmental characteristics.

Chapter 4 O-Cloud Requirements

This chapter describes design requirements for the O-Cloud. While most requirements apply to both the regional and edge O-Clouds, the edge O-Cloud may impose unique requirements, e.g., hardware accelerators, which will be noted specifically. In addition, depending on the implementation choices for O-CU, O-DU, and near-RT RIC, the O-Cloud needs to provide support for virtual machines or containers.

4.1 O-Cloud Architectural Components and Requirements

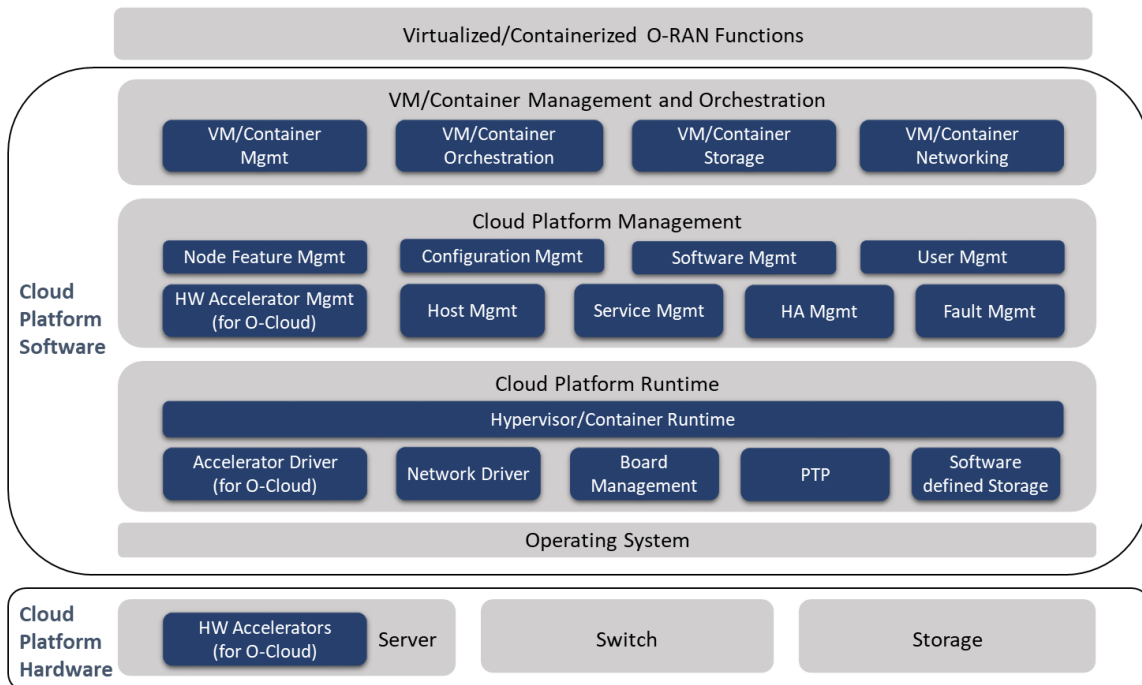


Figure 4-1. Architecture component requirements for the cloud platform.

Figure 4-1 depicts the high-level architecture components for deploying VM/container-based O-RAN functions in the regional and edge O-Clouds.

The Virtualized/Containerized O-RAN Functions include the O-DU, O-CU and other workloads deployed on the O-Cloud via the O2 interface.

The VM/Container Management and Orchestration is typically implemented with OpenStack for VM deployments and Kubernetes for container deployments.

The Cloud Platform Management represents the O-Cloud infrastructure management functions required for life cycle management, high availability, fault management and configuration of the O-Cloud platform.

The Cloud Platform Runtime represents the additional cloud platform run-time capabilities required to host O-RAN functions as compared to 5G core network functions.

4.1.1 Hardware Requirements

Table 4-1 shows requirements for the cloud platform hardware.

Components	Description
Server	- Regional cloud: standard COTS server - Edge cloud: depth $\leq 450\text{mm}$, height $\leq 2\text{RU}$, width $\leq 19\text{in}$ - Minimum 16-core CPU at 2.2 GHz base frequency and 128 GB DDR4 - At least 2x 25GbE ports for front-haul interface (or higher, depending on site), 2x 10GbE ports for backhaul interface, and 1x 1GbE out-of-band management port - SR-IOV-capable NICs with network link aggregation enabled - For single-socket, at least 3 PCI-E slots for 3x HHHL or 1x FHFL + 2x HHHL cards; for dual-socket, at least 4 slots for 4x HHHL or 1x FHFL DW + 3x LP cards - All hard disk drives require hot-pluggable - Management interfaces support IPMI, SNMP, and Redfish (preferred) - UEFI secure boot enabled
Hardware Accelerator	- Required for O-DU in the edge cloud in forms such as FPGA, ASIC, and GPU - At least 2 ports at 10 or 25 Gbps eCPRI/RoE open front-haul interface - Acceleration for functions such as LDPC encoding and decoding, and end-to-end high PHY
Switch	Standard COTS TOR switch
Storage	Software-defined storage (e.g., Ceph) based on COTS servers

Table 4-1. Requirements for the cloud platform hardware.

4.1.2 Operating System Requirements

Table 4-2 shows requirements for the cloud platform operating system.

Operating System	Description
Linux	- Linux kernel with real-time preemption patches for O-DU workloads (i.e., SMP PREEMPT RT). Real-time kernel patch for O-CU is optional. - Deterministic interrupt handling with a maximum latency of 20 μs as measured by cyclictst for system interrupts (e.g., external I/O) and interrupt-based O-CU and O-DU implementations (e.g., those that rely on OS timers) - CRI-O and/or CRI plugin containerd support - QEMU/KVM (or other types of hypervisor) for virtual machines

Table 4-2. Requirements for the cloud platform operating system.

4.1.3 Cloud Platform Runtime Requirements

Table 4-3 shows requirements for the cloud platform runtime.

Runtime Components	Description
Accelerator Driver	Edge cloud: driver for loading, configuring, managing and interfacing with accelerator hardware providing offload functions for O-DU container or VM

Crypto Driver	• Crypto offload driver for networking and O-CU wireless cipher (optional)
Network Driver	• Network driver(s) for front-haul, back-haul, mid-haul, inter container or VM communication, management and storage networks
Board Management	• Board management for interfacing with server hardware and sensors
PTP	• Precision time protocol for distributing phase, time and synchronization over a packet-based network
Software-defined Storage (SDS)	• Software implementation of block storage running on COTS servers, optional for edge cloud and required for regional cloud • If used in edge cloud, required to be hyperconverged; in regional cloud, either hyperconverged or deployed on dedicated storage nodes
Container Runtime	• Executes and manages container images on a node
Hypervisor	• Allows host to run multiple isolated VMs • Low performance overhead, compared with non-virtualized execution of specified benchmarks, e.g., SPECcpu, fio, L2Fwd, etc. (quantitative requirements FFS)

Table 4-3. Requirements for the cloud platform runtime.

4.1.4 Generic Requirements for Cloud Platform Management

Table 4-4 shows the generic requirements for management of the cloud platform.

Components	Description
Configuration Management	• Auto-discovery of new servers added to an O-Cloud • Management of installation parameters such as console, root disks • Configuration of core, memory, huge-page assignments on each server node • Network interfaces and storage assignments • Hardware discovery of CPUs, cores, storage, network ports, hardware accelerators, etc. • Virtual-to-physical CPU core and NUMA node pinning • Virtual NUMA topology configuration • Huge-page memory • Interrupt and thread pinning to virtual and physical cores
Host Management	• Life-cycle and availability management of the server • Host failure detection and recovery • Fault monitoring and reporting • Cluster connectivity failure detection and recovery • VM/container failure detection and recovery • Critical process failure detection and recovery • Resource utilization thresholds (e.g., CPU, memory, storage limits) • Network interface states • Hardware sensors and faults • Watchdog • Board management interfaces and out-of-band server management • Hardware sensor monitoring
Service Management	• Monitoring of critical platform infrastructure services and processes • Automatic recovery of failed platform infrastructure services and processes
HA Management	• High-availability services for supporting cloud platform redundancy
Fault management	• Cloud platform infrastructure alarm and fault management. • Ability to set, clear, query, filter, suppress, log and collect alarms
Software Management	• Ability to deploy software updates • Support for rolling updates across all nodes of the local O-Cloud • Support for migrating VMs and/or containers within the local O-Cloud for software upgrade procedures
User Management	• User authentication and authorization • Isolation of control and resources among different users
Node Feature Management	• Detection and setting of node-level policies to align resource allocation choices (i.e. NUMA, SR-IOV, CPU, etc.)

HW Accelerator Management	Support for managing hardware accelerators, mapping them to O-RAN applications VMs and/or containers, and updating accelerator firmware
---------------------------	---

Table 4-4. Cloud platform management requirements.

4.1.5 VM/Container Management and Orchestration Requirements

Table 4-5 shows requirements for VM/container management and orchestration as part of the cloud platform.

Components	Description
VM/Container Management	- VM management and scheduling with OpenStack Nova - Container management and scheduling with Kubernetes kube-apiserver, kube-scheduler, etcd, kube-controller-manager
VM/Container Orchestration	- VM orchestration with Heat templates - Container orchestration with HELM charts
VM/Container Storage	- VM storage and image services with OpenStack Glance and Cinder - Container persistent volume claims (PVCs) - VM/Container Ceph backends for persistent storage (optional)
VM/Container Networking	- VM networking with OpenStack Neutron - Container networking with kube-proxy and CNI

Table 4-5. Requirements for VM/container management and orchestration.

4.2 Time Synchronization Requirements

The synchronization requirements addressed in this document cover only the synchronization aspects of the O-Cloud nodes that are part of the edge cloud. O-Cloud nodes capable of supporting O-DU deployments shall comply with the synchronization requirements associated with the O-DU instances when instantiated on the O-Cloud nodes, as in CUS specification [2]. The requirements in the section address the LLS-C3 configuration where O-DU and O-RU are synchronized from the fronthaul network. Additional configurations are for future study.

This section assumes that the O-DU is not on the synchronization path to the O-RU, which is the case with the LLS-C3 and LLS-C4 configurations. O-RAN Fronthaul Working Group Control, User and Synchronization Specification (CUS) [2] specifies the time synchronization requirements for the LLS topology configurations.

O-DU synchronization requirements specified in the CUS [2] are more relaxed than the requirements associated with the O-RU. O-DU is able to meet synchronization requirements while selecting PTP source from connected networks, provided that the synchronization accuracy requirements for the network as specified in CUS [2] are met in such an implementation. As an end application, O-DU may optionally use SyncE along with PTP, or local time source (typically a GNSS receiver), but these options are not considered in this section. For more details on time synchronization requirements, refer to CUS [2].

4.2.1 G.8275.1 Full Timing Support (FTS)

Figure 4-2 illustrates a synchronization architecture for an edge site aligned with the CAD [1] LLS-C3 Synchronization Topology where Compute Nodes synchronize their clocks to a Grandmaster Clock via the Fronthaul Network.

LLS-C3 PTP Deployment Architecture

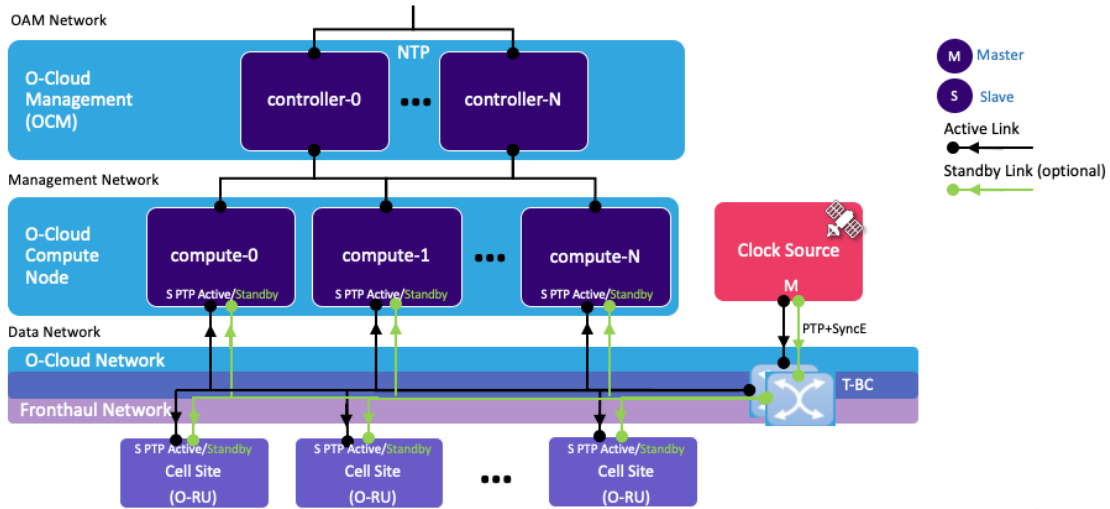


Figure 4-2. Key LLS-C3 Fronthaul Synchronization in an example implementation in O-Cloud.

A distinction is made between the O-Cloud network and the fronthaul network. The O-Cloud network includes all switches and routers in any network path between two O-Cloud nodes. The fronthaul network includes all switches and other transport equipment in any network path between an O-DU capable O-Cloud node and an O-RU. These networks may overlap, with some routers/switches (e.g., the leaf and spine switches in the O-Cloud resource pool) participating in both the O-Cloud network and the fronthaul network. In the case of such overlap, irrespective of the management responsibilities associated with fabric and transport network elements, all network elements in the timing transfer path between the Grandmaster Clock(s) and Slave (subordinate) Clocks are considered to be part of the fronthaul synchronization network from the associated synchronization functionality and profile compliance point of view.

An O-Cloud synchronized from the fronthaul network, based on full time support, shall follow the following requirements:

O-Cloud node requirements:

- (1) All O-DU capable O-Cloud nodes shall have at-least one NIC connected to the fronthaul network. Such nodes shall implement PTP Slaves (subordinate) to the fronthaul network that provides the PTP timing synchronization.
- (2) The PTP Slave (subordinate) NICs shall support T-TSC functionality that is fully compliant with PTP profile ITU-T G.8275.1.
- (3) These Slave (subordinate) PTP NICs shall implement according to “1588-2008 - IEEE Standard for a Precision Clock Synchronization Protocol for Networked Measurement and Control Systems”, section “6.6.5 Generation of PTP message timestamps” reference point A.
- (4) The O-Cloud shall track which nodes supports PTP capable NICs that are connected to the fronthaul network for the purpose of inventorying O-Cloud nodes that satisfy vO-DU synchronization requirements.
- (5) The O-Cloud shall provide the ability to configure an O-Cloud node’s PTP physical port(s) that receives PTP traffic as a PTP Slave (subordinate). The following minimum configuration shall be supported:
 - Network Transport – G.8275.1 synchronization over Ethernet (Layer 2);
 - Time Stamping – support for hardware time stamping.
- (6) The O-Cloud operating environment (software stack) shall support PTP protocol implementation (1588-2008 - IEEE Standard). For example: in the case when an O-Cloud is based on the Linux OS, this will require support for ptp4l and phc2sys.
- (7) The node clock on each node shall be synchronized from the NIC T-TSC clock(s). For example: in the case when an O-Cloud is based on the Linux OS, this will require support for phc2sys – Synchronization of two clocks, PHC and node clock (a.k.a system clock) when using HW timestamping.
- (8) Administrative access to the PHCs from deployment instances shall not be allowed, as that would interfere with the PHC management from the cloud platform software. The O-Cloud may provide direct read-only access to the PHC counters from the deployment instances.
- (9) When the Slave (subordinate) PTP NIC is connected two at least two ports to the fronthaul network the PTP Slave (subordinate) NIC shall run the Alternative BMCA in order to select its best master clock.

O-Cloud network requirements:

- (1) All O-Cloud switches that are also part of the fronthaul network shall provide full timing support with the G.8275.1 Telecom Profile. The synchronization performance requirements are specified in the CUS specification [2] for the O-DU LLS-C3 configuration. Although the CUS allows the ITU-T G8275.2 Telecom Profile for partial timing support, O-Cloud requirements and performance with G8275.2 is for future study.
- (2) Time synchronization redundancy should be supported and is up to the operator's discretion.
- (3) O-Cloud network switches that are also part of the fronthaul network shall support untagged L2 Ethernet transport for S-Plane traffic.
- (4) The fronthaul network elements between the GM and RUs shall require implementing SyncE based on O-RU requirements.

Recommendations:

All physical network elements that are in the synchronization transfer path (i.e. PRTC/T-GM, switches that act as T-BC, and all O-Cloud node Slaves (subordinate) that act as T-TSC) should implement a 1PPS output interface of the associated Network Elements local clock to facilitate connection to external performance test equipment for performance validation purposes.

Note: If the O-RUs are synchronized from the same clock(s), which is expected to be a typical case, O-RU rather than O-DU related synchronization requirements determine the performance classes of the PRTC/T-GM and T-BCs that are on the synchronization distribution path towards the O-RUs.

Note: The Alternative Best Master Selection Algorithm (BMCA) running in Ordinary Clock or Boundary Clock performs the GM selection. A-BMCA is used for improving the accuracy, to optimize the time distribution with multiple PRTC/T-GMs and providing redundancy. By default, the BMCA runs at least once every Announce message interval. There are two steps for the selection:

- Data Set Comparison Algorithm to select better source from Grandmaster. Data Sets are compared from the Announce message from Multiple Sources.
- State Selection Update the state of each port based on the selection. This is to set the state of all the ports in the clock to a new state, if needed, to sync with better source of a GM.

4.2.2 G.8275.2 Partial Timing Support (PTS)

As specified in the CUS specification [2], DU synchronization requirements are more relaxed than the requirements associated with the RU, and CUS specification [2] also allows separate PRTC sources to be used for the synchronization of the RUs and DUs. The use of G.8275.2 Partial Timing Support is allowed for the synchronization of O-Cloud nodes (and instantiated O-DU's), provided that the synchronization accuracy requirements for the O-DU as specified in CUS specification [2] are met in such implementation. The detailed implementation of this configuration is subject for the further study.

4.3 Hardware Acceleration Requirements

The O-RAN Cloud Architecture document [1] discusses the hardware abstraction considerations for the O-Cloud. It lists two levels of abstraction referring to the deployment model and APIs, and also discusses the management and orchestration considerations that should be supported by such hardware accelerators deployed on the cloud platform. This section describes the types of acceleration as background information and then specifies requirements applicable to the O-Cloud for acceleration hardware abstraction.

4.3.1 Background on Acceleration Types

Hardware acceleration for the managed functions, such as O-DU and O-CU, can be classified into two types:

- Interface acceleration: acceleration of I/O data transfers on any interface between 3GPP or O-RAN network functions [16];
- Algorithm acceleration: acceleration of any algorithm processing within an O-RAN network function.

Interface acceleration mostly exhibits as acceleration of network traffic transmission and reception, for which SR-IOV NICs, combined with software mechanisms such as DPDK, is commonly used.

Algorithm acceleration includes acceleration of compute-intensive algorithms in the O-DU and O-CU, utilizing accelerator hardware such as FPGA, GPU, DSP, and ASIC.

4.3.1.1 Acceleration for the O-DU

Figure 4-3 illustrates acceleration of the interfaces between the O-DU and other O-RAN network functions and acceleration of the high-PHY algorithms within the O-DU. Table 4-6 shows in more detail the accelerated functions.

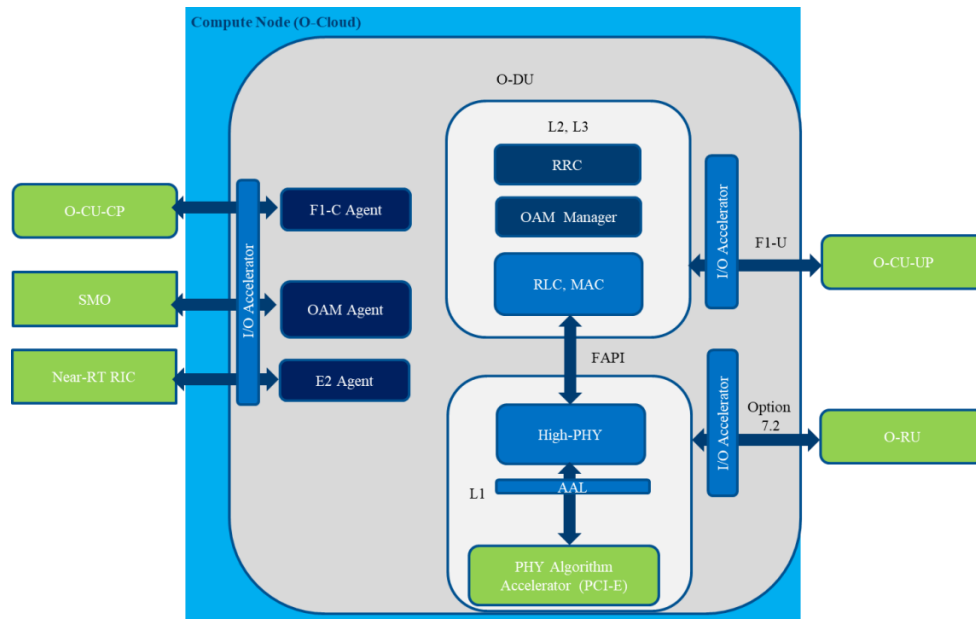


Figure 4-3. O-DU interface and algorithm acceleration.

Accelerated Function	Interface Name	Acceleration Type
I/O between O-DU and O-CU-CP	F1-C	Interface acceleration (optional)
I/O between O-DU and O-CU-UP	F1-U	Interface acceleration
I/O between O-DU and SMO	O1	Interface acceleration (optional)
I/O between O-DU and Near-RT RIC	E2	Interface acceleration (optional)
I/O between O-DU and O-RU	FH7.2	Interface acceleration
High-PHY algorithms	N/A	Algorithm acceleration

Table 4-6. O-DU acceleration types.

4.3.1.2 Acceleration for the O-CU-CP

Figure 4-4 illustrates acceleration of the interfaces between the O-CU-CP and other O-RAN network functions and acceleration of the security algorithms within the O-CU-CP. Table 4-7 shows in more detail the accelerated functions.

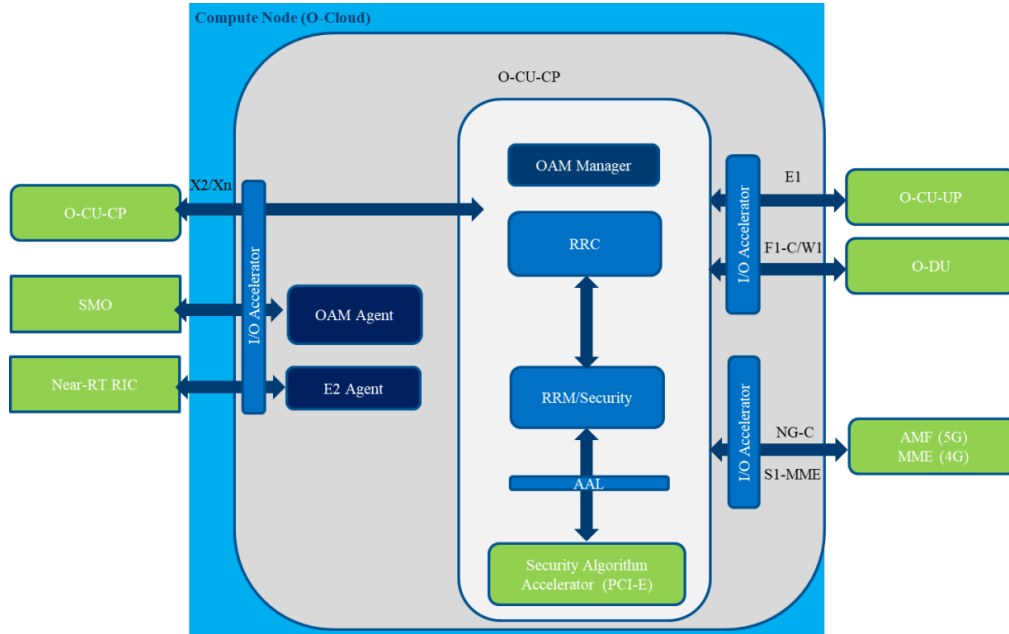


Figure 4-4. O-CU-CP interface and algorithm acceleration.

Accelerated Function	Interface Name	Acceleration Type
I/O between O-CU-CP and O-DU	F1-C	Interface acceleration (optional)
I/O between O-CU-CP and SMO	O1	Interface acceleration (optional)
I/O between O-CU-CP and Near-RT RIC	E2	Interface acceleration (optional)
I/O between O-CU-CP and O-CU-UP	E1	Interface acceleration (optional)
I/O between O-CU-CP and O-CU-CP	Xn	Interface acceleration (optional)
I/O between O-CU-CP and O-CU	X2	Interface acceleration (optional)
I/O between O-CU-CP and MME	S1AP	Interface acceleration (optional)
I/O between O-CU-CP and AMF	NG-C	Interface acceleration (optional)
Security algorithms	N/A	Algorithm acceleration

Table 4-7. O-CU-CP acceleration types.

4.3.1.3 Acceleration for the O-CU-UP

Figure 4-5 illustrates acceleration of the interfaces between the O-CU-UP and other O-RAN network functions and acceleration of the security algorithms within the O-CU-UP. Table 4-8 shows in more detail the accelerated functions.

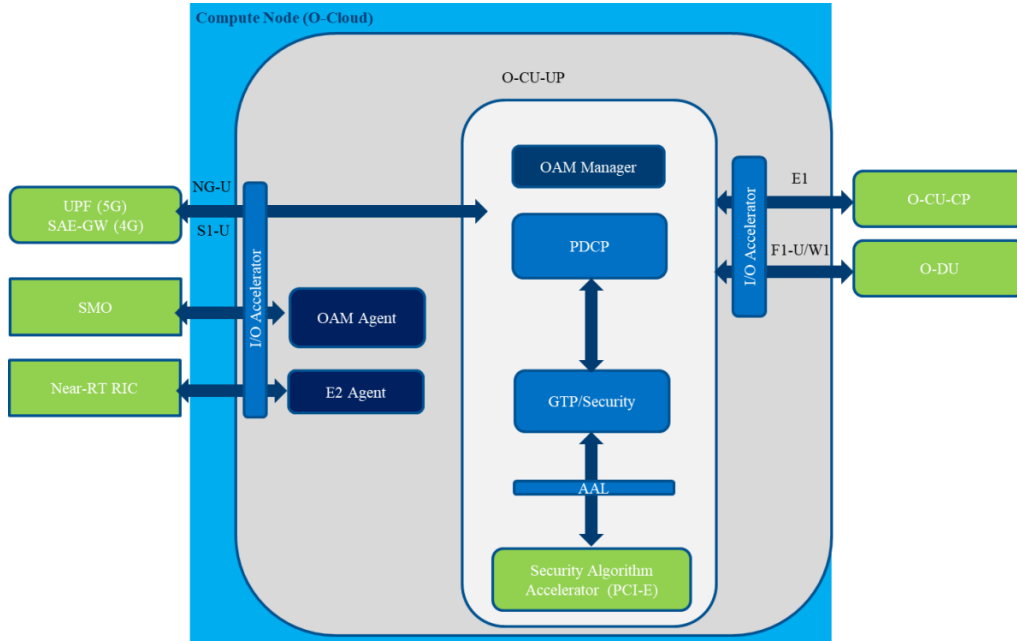


Figure 4-5. O-CU-UP interface and algorithm acceleration.

Accelerated Function	Interface Name	Acceleration Type
I/O between O-CU-UP and O-DU	F1-U	Interface acceleration
I/O between O-CU-UP and SMO	O1	Interface acceleration (optional)
I/O between O-CU-UP and Near-RT RIC	E2	Interface acceleration (optional)
I/O between O-CU-UP and O-CU-CP	E1	Interface acceleration (optional)
I/O between O-CU-UP and SAE-GW	S1U	Interface acceleration
I/O between O-CU-UP and UPF	NG-U	Interface acceleration
Security algorithms	N/A	Algorithm acceleration

Table 4-8. O-CU-UP acceleration types.

4.3.2 Hardware Acceleration Abstraction Layer Requirements

The Acceleration Abstraction Layer (AAL) provides a set of APIs to VNF/CNF software for offloading certain functions that, for instance, are compute- and power-intensive, to hardware accelerators supported on the O-Cloud platform. The objective of the AAL is to hide the different implementations of hardware accelerators and present APIs to VNF/CNF applications to enable portability and software-and-hardware decoupling. Full listing of the functions that can be offloaded through the AAL is outside the scope of this document.

4.3.2.1 Generic Requirements

Extensibility

O-RAN has defined the functions that can be accelerated by the cloud platform based on 3GPP specifications and O-RAN deployment scenarios. However, the AAL API should not limit innovation of future implementations and should evolve as the specification requires. In this way, the API shall be extensible to accommodate future revisions of the specification.

Interrupt and Poll Mode

The API shall allow multiple design choices for application vendors and shall not preclude a vendor from adopting an interrupt-driven design or poll-mode design or any combination of both. As such, the API shall support both interrupt mode and poll mode for the data-path application interface.

Discovery and Configuration

The API shall support application software to discover and configure the accelerator hardware. The API shall allow an application to discover what physical resources have been assigned to it from the upper layers and then to configure said resources for offload operations.

Multiple Device Support

There may be scenarios where multiple hardware accelerators (either implementing the same acceleration function or different) are assigned to a single application, which uses one or more of these accelerators as needed. The API shall support an application using one or more accelerator devices at the same time, as Figure 4-6 illustrates.

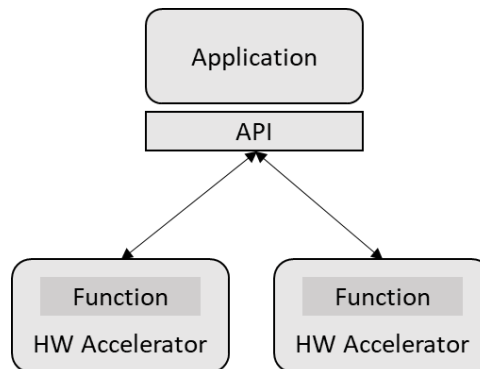


Figure 4-6. AAL support for multiple devices.

4.3.2.2 API offload requirements

The API in supporting different implementations shall support different offload architectures including look-aside, inline, and any combination of both.

Look-aside Acceleration Model

The API shall support look-aside acceleration model where the host CPU invokes an accelerator for data processing and receives the result after processing is complete. A look-aside architecture, illustrated in Figure 4-7, allows the application to offload work to a hardware accelerator and continue to perform other work in parallel—this could be to continue to execute other software tasks in parallel or to sleep and wait for the accelerator hardware to complete. This model requires the API to support two operations, one for initiating the offload and another for retrieving the operation once complete.

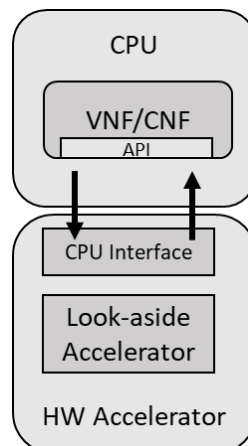


Figure 4-7. AAL look-aside acceleration model.

Inline Acceleration Model

The API shall support inline acceleration model where acceleration by function and I/O-based acceleration are performed on the physical interface as the packet ingresses/egresses the platform. Figure 4-8 shows one possible implementation of an inline acceleration model.

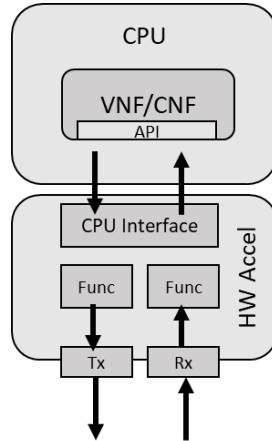


Figure 4-8. AAL inline acceleration model.

In Figure 4-8, a “Func” block refers to some algorithmic function that is accelerated in the device before the data egresses or ingresses from the acceleration device. “Tx” refers to the transmission of the data from the acceleration device to an Ethernet interface, while ‘Rx’ refers to the reception of data from the Ethernet interface to the acceleration device.

While the look-aside architecture shall support dataflow from the CPU to the accelerator and back to the CPU before being sent to the front-haul interface, the inline architecture shall support data flow from the CPU to the accelerator and directly from the accelerator to the front-haul interface, instead of being sent back to the CPU. The typical data flows for accelerating the O-DU high-PHY functions for the look-aside and inline architectures are as follows.

Look-aside architecture dataflow

CPU ↔ accelerator ↔ CPU ↔ front-haul: for a set of consecutive PHY functions offload (e.g., FEC)

CPU ↔ accelerator ↔ CPU ↔ accelerator ↔ ... ↔ CPU ↔ front-haul: for a set of non-consecutive PHY functions offload

Inline architecture dataflow

CPU ↔ accelerator ↔ front-haul: for a set of consecutive PHY functions offload (up to the end of the PHY pipeline)

Figure 4-9 illustrates one possible implementation of the look-aside and inline architectures. While a set of PHY-layer functions are offloaded to the accelerator hardware for look-aside acceleration, the entire end-to-end high PHY pipeline is offloaded to the accelerator for inline acceleration.

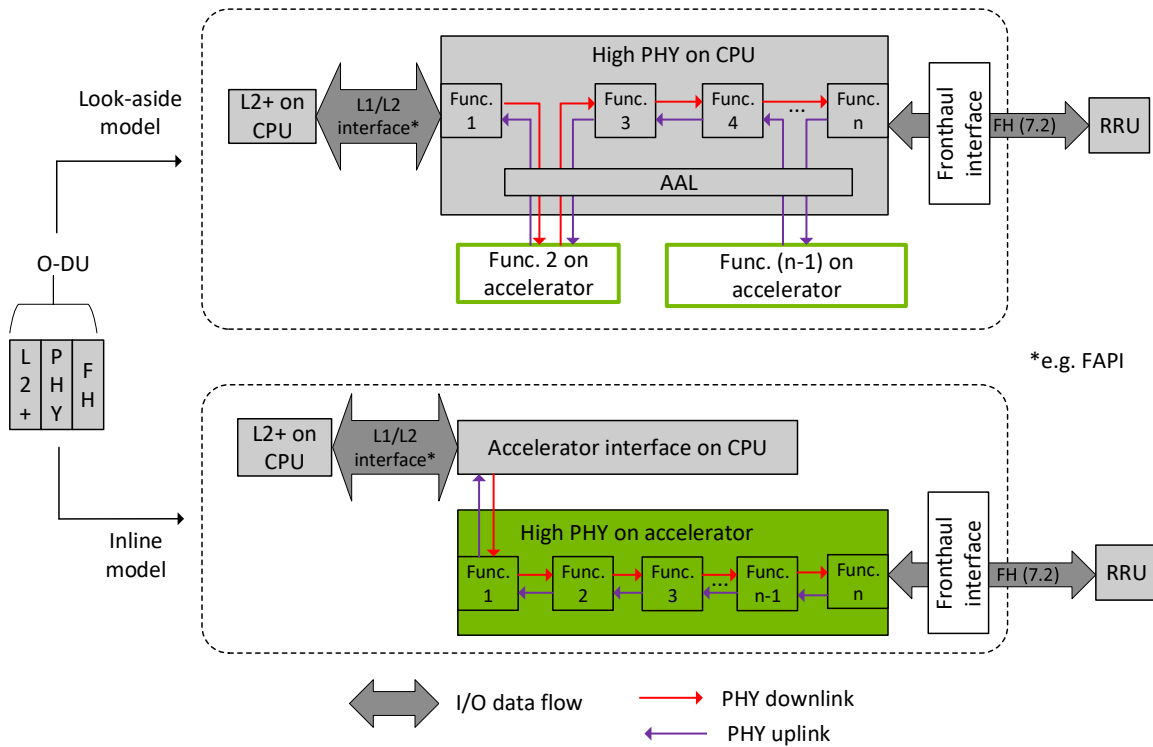


Figure 4-9. Dataflow paths in look-aside and inline acceleration architectures.

API Concurrency and Parallelism

To enable greater flexibility and design choice by application vendors, the API shall support multi-threading environment allowing an application to offload acceleration requests in parallel from several threads.

Chapter 5 (Informative) Cloud Platform Design Example

This chapter includes some possible platform configurations for the regional and edge clouds for both the cloud platform hardware and software. These platform configurations are informative examples only, not normative, and focus on sub-6GHz scenarios. Support for mmWave and quantification of capacities of these example designs (such as the number of cells, data rates, and so forth) have not been systemized or validated and will be further studied in the next release of this document.

5.1 Cloud Platform Hardware Design Example

This section specifies the cloud platform hardware with a focus on server and accelerator hardware. These informative designs are some examples only. This document focuses on x86 processor-based cloud server design; it will address ARM processor-based designs in a future release.

5.1.1 Regional Cloud Server Example

Table 5-1 shows an example design for the regional cloud servers.

Processor	2x 20-core CPUs at 2.4 GHz, e.g., 2 nd -generation Intel Xeon Scalable Processor
Memory	12x 32 GB, 2400 MHz TruDDR4
Expansion Slots	6x PCIe 3.0
Drive Bays	14x 3.5" or 24x 2.5" hot-swap bays or 24x NVMe bays

RAID support	RAID 1, 12 Gb/s
Network Interface	4x 10GbE/25GbE ports, 1 dedicated 1GbE management port, e.g., 1x Fortville 40GbE Ethernet PCIe XL710-QDA2 Dual Port QSFP+ (PCI-E card direct to CPU 0)
Network link aggregation	Enabled
Power	2x hot swap/redundant; support 220V/380V AC, 480V and -48V DC
Security	TPM or TCM
Availability	Hot-swap fans/PSUs; 45°C continuous operation; light path diagnostic LEDs; front-access diagnostics via dedicated USB port
Manageability	Support for IPMI, SNMP, and Redfish

Table 5-1. Regional cloud server example design.

5.1.2 Edge Cloud Server Example

Table 5-2 shows an example design for the edge cloud servers with FPGA based acceleration.

Form factors	Depth $\leq$ 450mm, height $\leq$ 2RU, width $\leq$ 19in
Processor	2 nd -generation Intel Xeon Gold (e.g., 6210U) or Xeon D-2187NT (16-core) or Xeon D-2177NT (14-core); hyperthreading enabled for O-CU, disabled for O-DU Note: the NT models have Intel QAT integrated with the CPU and can perform PDCP crypto acceleration in the vO-CU (when applicable, i.e., for Scenario B and C.1/C2).
Memory	12x 32 GB, 2400 MHz TruDDR4
Expansion Slots	At least 2x half-height half-length (HHHL) or 1x HHHL and 1x FHFL
Drive Bays	6x hot-swappable 2.5-inch SATA or SAS drives; 2x NVMe drives; all drives accessible from front panel
RAID support	RAID1
Network Interface	4x 10GbE/25GbE ports, 1 dedicated 1GbE management port, 1x Fortville 40GbE Ethernet PCIe XL710-QDA2 Dual Port QSFP+ (PCI-E card direct to CPU 0), network cabling at front panel
PSU	1+1 redundant - Hot-swappable - Support for 220V AC or -48V DC - PSUs are connected through the rear panel (AT&T, China Mobile) or front panel (China Telecom, DT, KDDI, Orange)
Temperature	-5 ° C ~ 55 ° C (-10 ° C to 60 ° C preferred)
Security	TPM or TCM
Availability	Hot-swap fans/PSUs; 45°C continuous operation; light path diagnostic LEDs; front-access diagnostics via dedicated USB port
Manageability	Support for IPMI, SNMP, and Redfish
Acceleration (look-aside, FEC profile)	Intel® Programmable Acceleration Card N3000, PCIe x16 Gen3, FHHL, I/O 2x25Gbe per QSFP28, FEC Profile 15Gbps Uplink, 5Gbps Downlink, Up to 8 VFs.

Table 5-2. Edge cloud server example design with FPGA acceleration.

Table 5-3 shows an example design for the edge cloud servers with GPU based acceleration.

Form factors	Depth $\leq$ 450mm, height $\leq$ 2RU, width $\leq$ 19in
--------------	--

Processor	12C x86 CPU complex or better (e.g. Xeon)
Memory	64 GB, 2400 MHz; or higher
Expansion Slots	At least 1x HHHL and 1x FHFL Dual Slot
Drive Bays	6x hot-swappable 2.5-inch SATA or SAS drives; 2x NVMe drives; all drives accessible from front panel
RAID support	RAID1
Network Interface	2x 100/50/25/10 GbE ports, 1 dedicated 1GbE management port, 1x Mellanox ConnectX®-6 Dx (with crypto offload support, eCPRI offload with 5T for 5G technology) Efficient peer-to-peer support, either through root complex, or external PCIe switch (such as ExpressLane™ PEX 8747) recommended for high bandwidth use cases (like multi-carrier 100MHz massive MIMO systems)
PSU	1+1 redundant - Hot-swappable - Support for 220V AC or -48V DC - PSUs are connected through the rear panel (AT&T, China Mobile) or front panel (China Telecom, DT, KDDI, Orange)
Temperature	-5 ° C ~ 55 ° C (-10 ° C to 60 ° C preferred)
Security	TPM
Availability	Hot-swap fans/PSUs; 45°C continuous operation; light path diagnostic LEDs; front-access diagnostics via dedicated USB port
Manageability	Support for IPMI, SNMP, and Redfish
Acceleration (inline, end-to-end HIGH PHY profile)	NVIDIA® T4, PCIe x16/x8 Gen3, HHHL (for smaller capacity, low band deployments). NVIDIA® Tesla® V100, PCIe x16 Gen3, FHFL (for higher capacity, to enable support for multi-sector massive MIMO (100 MHz and above) with advanced beam-forming techniques).

Table 5-3. Edge cloud server example design with GPU acceleration.

5.1.3 BIOS/UEFI Configurations

Table 5-4 shows recommended settings for the various edge cloud deployment types.

BIOS Setting	Description	Regional Cloud Settings for Maximum Performance (with Turbo mode enabled)	Regional & Edge Cloud with Real Time	Edge Cloud with Strict Real Time Configuration for O-DU
System Power and Performance Policy	Favor high performance over low power consumption for the entire system (e.g., CPU and memory).	Optional	Enabled	Enabled
Intel Hyper-Threading Technology (HT)	Enabling HT doubles the number of (logical) CPU cores. But threads on the same physical core share various core data structures and caches, causing performance interference. It is recommended to enable HT to ensure high system throughput, but for applications such as O-DU in the edge cloud, carefully pin	Enabled	Optional	Optional

	latency-sensitive threads to dedicated cores to ensure real-time performance.			
Prefetching	Memory cache prefetching technologies allow prefetching data and instructions into the cache before they are used. In general, prefetching reduces cache misses and increases application performance.	Enabled	Enabled	Enabled
Intel virtualization support (VT-x)	CPU hardware support for virtualization.	Enabled	Enabled	Enabled
Intel VT for direct I/O (VT-d)	Support for PCI-E passthrough and SRIOV.	Enabled	Enabled	Enabled
Intel Turbo Boost Technology	Enabling this feature allows individual CPU cores to run at frequencies higher than the base frequency when the processor has sufficient thermal headroom. Frequent transition between normal and turbo modes, however, can cause non-predictive performance and impact system real-time characteristics.	Enabled	Disabled	Enabled
Enhanced Intel SpeedStep Technology	Enhanced SpeedStep is a series of dynamic frequency scaling technologies built into modern Intel microprocessors that allow the clock speed of the processor to be dynamically changed (to different P-states) by software. This allows the processor to meet instantaneous performance needs of the operation being performed, while minimizing power draw and heat generation. In general, disabling this feature avoids CPU frequency changing dynamically and thus facilitates deterministic performance. Enabling this but setting the configurable TDP level in the next setting allows finer control on the frequency that the CPU will run at.	Enabled	Disabled	Enabled
Intel Configurable TDP		Nominal	Nominal	Level 2
Energy-efficient Turbo	Transitions in and out of turbo mode incur extra power and overly frequent transitions are energy inefficient. This feature enables more energy-efficient turbo operation.	Disabled	Disabled	Disabled
Hardware P-States	Disabling hardware-controlled P-states avoids unpredicted frequency change and helps achieve deterministic performance.	Disabled	C0/C1 state	C0/C1 state
Hardware PM Interrupt		Disabled	Disabled	Disabled
EPP Enable		Disabled	Disabled	Disabled
APS Rocketing		Disabled	Disabled	Disabled
Scalability		Disabled	Disabled	Disabled
PP0-Budget		Disabled	Disabled	Disabled

CPU C State Control & C1E & Processor C6	Disabling C-state transitions prevents CPU cores from entering sleep states when there is no task to execute. C-states that are often seen in BIOS settings include C1E, C3, and C6 states. As C-state transitions introduce delay, disabling them helps real-time performance, though at the cost of higher power consumption.	Enabled	Disabled	Disabled
Direct Cache Access (DCA) & Data Direct I/O (DDIO)	DCA can only reduce packet reception delay. Another technology, DDIO, reduces delay for both transmission and reception. It is recommended to enable DDIO instead of DCA when both settings are present.	Enabled	Enabled	Enabled
Uncore Frequency Scaling	Allow the voltage and frequency of the uncore to be programmed independently. Uncore activities are monitored to optimize the frequency in real time.	Enabled	Disabled	Enabled
Memory RAS and Performance Configuration & NUMA Optimized	Enable ACPI tables that are required for NUMA-aware operating systems.	Optional	Optional	Optional

Table 5-4. BIOS configurations for strong real-time systems.

5.1.4 Firmware Versions

Table 5-5 shows firmware versions for the example reference design based on the Intel Server Board S2600WF family.

BIOS/IFWI	SE5C620.86B.00.01.0013.030920180427
Baseboard Management Controller (BMC)	1.43.91f76955
Download link	https://downloadcenter.intel.com/download/27632/Intel-Server-Board-S2600WF-Family-BIOS-and-Firmware-Update-Package-for-UEFI-?product=89005

Table 5-5. Firmware for example reference design based on Intel server board S2600WF.

5.1.5 Accelerator Hardware

For deployment scenario B (see [1] and Section 2.1), the O-DU and the O-CU functions are deployed as VNF(s) or CNF(s) on the edge cloud platform and the Near-RT RIC is deployed as a VNF or CNF on the regional cloud platform. Likewise, for deployment scenario C and its variants C.1/C.2 (see [1] and Section 2.2), the O-CU functions are also deployed as VNF(s) or CNF(s) on the regional cloud platform. This section details the hardware acceleration options for these VNFs/CNFs that the cloud platforms shall provide.

5.1.5.1 Regional Cloud Platform

In deployment scenario B, the regional cloud platform provides support for the Near-RT RIC. There is currently no hardware acceleration options required for the Near-RT RIC at this stage of definition but this may evolve in future.

However, in deployment scenario C, the regional cloud platform also provides support for the O-CU in addition to the Near-RT RIC. The O-CU, depending on performance requirements, may choose to use hardware acceleration (e.g., Intel QAT) for wireless cipher processing, namely 128-EEA3, 128-EIA3, UEA2 and UIA2—depending on configurations.

5.1.5.2 Edge Cloud Platform

The O-CU and O-DU may follow the software architecture as defined in the O-RAN WG8 specification [17].

O-CU Acceleration

When the O-CU functions are deployed on the edge cloud platform (i.e., scenarios B and C.1/C2), the O-CU, depending on performance requirements, may choose to use hardware acceleration (e.g., Intel QAT) for wireless cipher processing, namely 128-EEA3, 128-EIA3, UEA2 and UIA2—depending on configurations.

O-DU Acceleration

The O-DU is assumed to use the 7.2 split with the front-end processing (FFT, PRACH) implemented in the O-RU. The remaining O-DU functions are implemented as software running on the edge cloud platform. Depending on the performance requirements, the compute-intensive workloads may be accelerated in hardware using look-aside acceleration or inline acceleration or any combination of both.

Figure 5-1 shows a high-level architecture of look-aside acceleration with proposed Forward Error Correcting (FEC) functions to be hardware-accelerated for the O-DU.

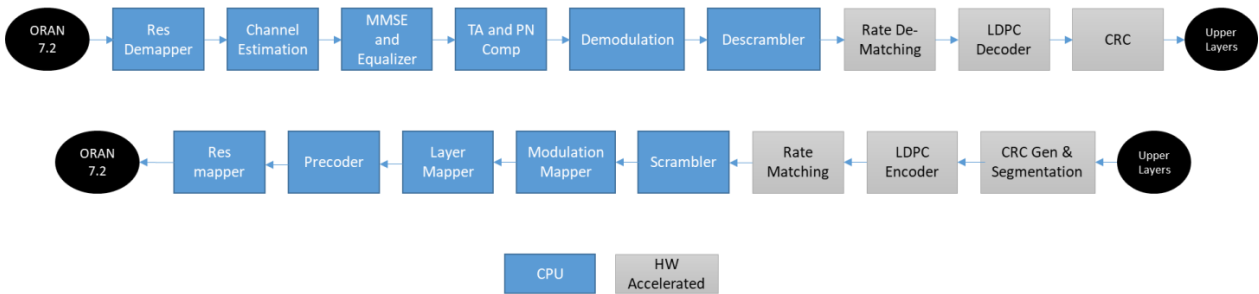


Figure 5-1. Hardware-accelerated FEC functions with look-aside model.

Figure 5-2 shows an example of an inline solution for the O-DU that accelerates the front-end compression and decompression (e.g., Mu-Law and BFP) functions in accelerator hardware before sending to the O-RU.

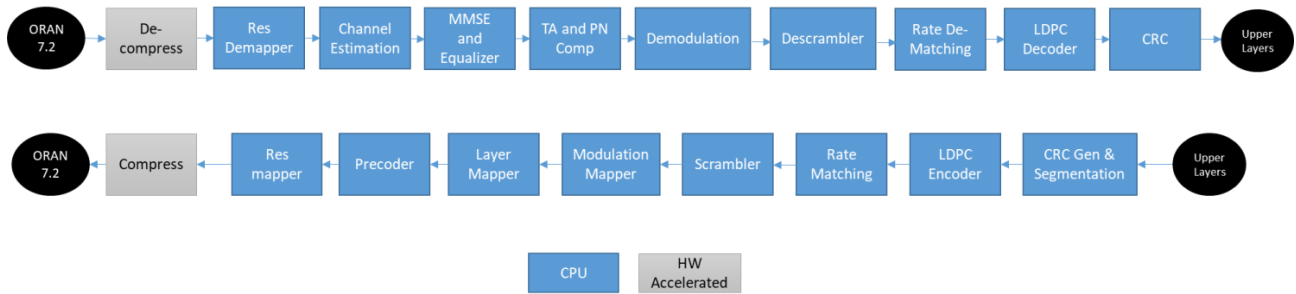


Figure 5-2. Hardware-accelerated front-end functions with inline model.

Figure 5-3 shows a high-level architecture of inline acceleration with end-to-end high PHY functions to be hardware-accelerated.

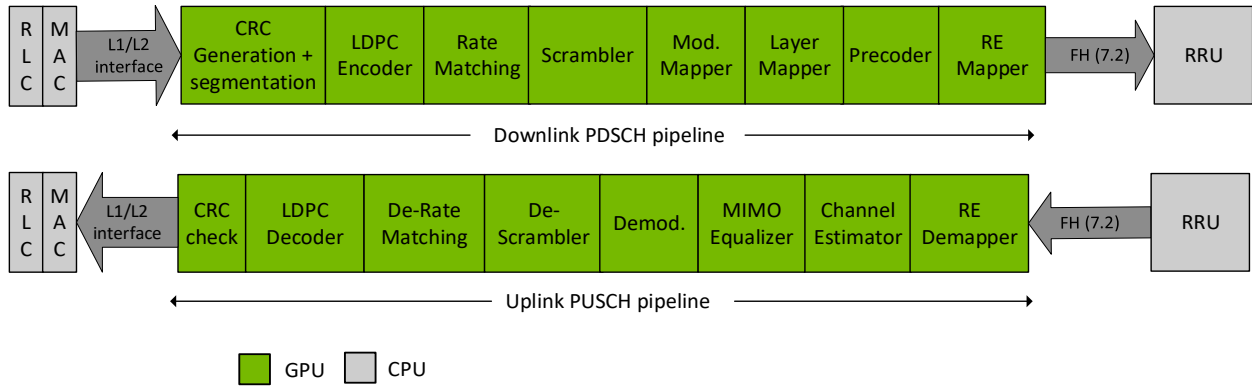


Figure 5-3. Hardware-accelerated end-to-end high-PHY functions with inline model.

5.1.5.3 High-level Platform Architecture Examples

Based on the hardware acceleration options above, Figure 5-4 to Figure 5-6 illustrate the system-level platform components required for the cloud platform to support various acceleration architectures for the O-DU and the O-CU, and depicting also both scenarios B and C.

A cloud platform can also support a combination of accelerator functions, such as inline front-end acceleration and FEC look-aside acceleration. Note that hardware acceleration for O-RAN network functions is desired but a software-only implementation is also possible.

Example 1: O-DU and O-CU look-aside acceleration

Figure 5-4 shows an example cloud platform and its data flows with look-aside acceleration for both the O-DU (FEC) and O-CU (wireless cipher).

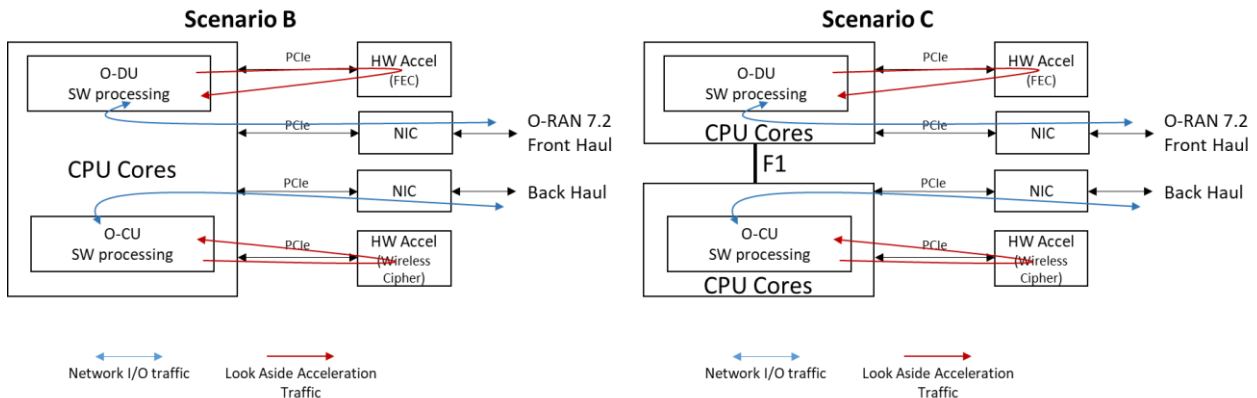


Figure 5-4. Cloud platform for look-aside acceleration of O-DU and O-CU.

Example 2: O-DU inline and O-CU look-aside acceleration

Figure 5-5 shows an example cloud platform with mixed acceleration models, where the O-DU uses the inline model for front-haul processing and the O-CU adopts look-aside acceleration for wireless ciphering.

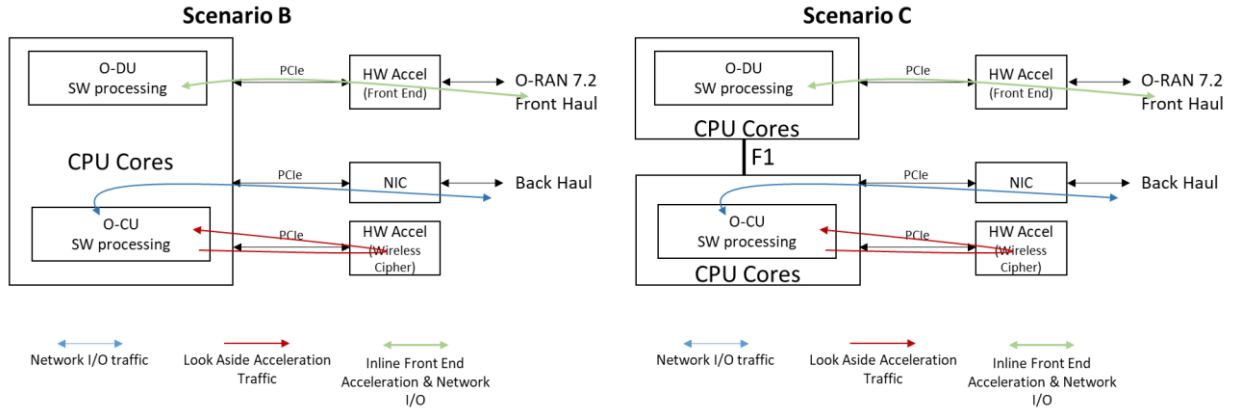


Figure 5-5. Cloud platform for O-DU inline acceleration and O-CU look-aside acceleration.

Example 3: O-DU end-to-end high-PHY inline acceleration and O-CU look-aside acceleration.

Figure 5-6 shows an example cloud platform, where the O-DU uses the inline model for the entire high-PHY processing and the O-CU adopts look-aside acceleration for wireless ciphering.

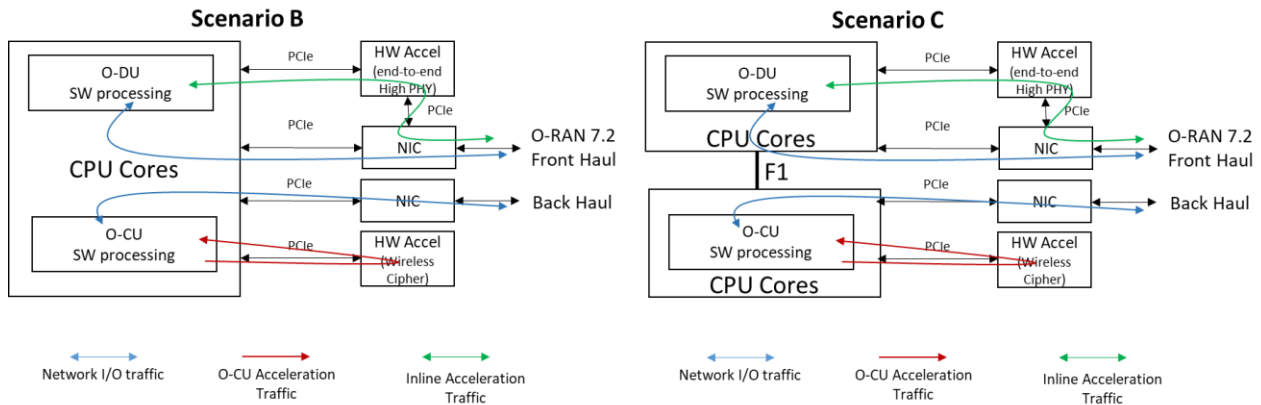


Figure 5-6. Cloud platform for O-DU inline acceleration of end-to-end high-PHY.

5.2 Cloud Platform Software Reference Design

This section shows an example reference design for the regional and edge cloud platform software. A single installable package supporting various profiles of the platform software runs in both the regional and edge clouds, and can adjust to the different physical environments and scale requirements. As mentioned in Chapter 2, this document focuses on a container-based cloud platform design and will address VM-based designs in a future release.

Kubernetes is a common implementation for container orchestration and management. Platform infrastructure management and additional components, such as Kubernetes plugins, must be added for hosting accelerated O-DU and O-CU network functions.

The regional and edge cloud platform requirements also differ based on the network functions they host. Generally, the edge cloud platform requirements are a superset of the functional requirements of the regional cloud platform. The primary differences relate to hardware acceleration features, real-time Linux, precision time protocol and server node scalability down to a single node for cell-site deployments.

651 This reference design focuses on deployment scenario B and C (with its variants C.1/C.2), LLS-C3 clock synchronization,
652 and FPGA and GPU hardware acceleration at the edge cloud. It defines two reference configuration profiles, for the
653 regional cloud and edge cloud.

654 5.2.1 Common Reference Configurations for all Profiles

655 Table 5-6 lists the primary platform software components that are common across the two cloud platform reference
656 configuration profiles. A complete list of packages, version information and build recipes to create a cloud platform
657 meeting the requirements of Section 4.1.3 for container deployment are available in O-RAN SC INF project release B
658 (Bronze release). See [18] for further details.

Platform Components	Description
Ansible	Orchestration engine for automating configuration management and application deployment. ansible.com
Ceph	Provides object storage, block storage and POSIX-compliant network file system. See ceph.io .
Collectd	Collects system and application performance metrics and provides mechanisms to store values. collectd.org
Dhcp	Assigns IP address and other network information to hosts.
Dnsmasq	Provides Domain Name System forwarder. en.wikipedia.org/wiki/Dnsmasq
docker-distribution	Docker registry for storing and distributing images. github.com/docker/distribution
Drbd	Replicated storage.
Etc	Distributed key-value store. etcd.io
Helm	Kubernetes package manager. helm.sh
helm-charts	Tool for managing charts. Charts are packages of pre-configured Kubernetes resources. github.com/helm/charts
ima	Detects if files have been accidentally or maliciously altered, both remotely and locally. linux-ima.sourceforge.net
iptables	Configures the tables from the Linux kernel firewall and the chains and rules it sores. en.wikipedia.org/wiki/Iptables
Kubernetes	Automates deployment, scaling and management of containerized applications. kubernetes.io
mariadb	Relational databases. mariadb.org
nginx	High-performance HTTP server, reverse proxy and load balancer.

pxe-network-installer	Provides PXE (preboot execution environment) install server.
rabbit-mq	Highly-available and scalable message broker. rabbitmq.com
redfishtool	Industry standard protocol providing a RESTful interface for the management of servers.
tpm2	Provides trusted platform module support.
Kubernetes Components	Description
etcd	Distributed key-value store used as Kubernetes backing store for cluster data. etcd.io
coredns	Kubernetes cluster DNS service.
armada	Manages helm chart dependences.
calico	Enables networking and network policy in Kubernetes clusters. Calico uses a pure IP networking fabric to provide high-performance networking.
nginx-ingress-controller	Enables Kubernetes to configure nginx for load balancing Kubernetes based services
tiller	Service that communicates with the Kubernetes API to manage helm packages.

Table 5-6. Common cloud platform reference configurations for all profiles.

5.2.2 Regional Cloud Reference Configuration Profile

The regional edge cloud profile is a cluster-based implementation with two redundant cloud platform controller nodes and worker node scalability from one to 200 nodes. In addition to providing a cloud platform for the Near-RT RIC and, additionally for the O-CU in Scenario C, the regional cloud platform provides zero-touch deployment and provisioning and management of the edge clouds.

The SMO interfaces to the regional cloud platform for deployment of new edge clouds as well as updating and upgrading of the edge cloud platform software. The regional controller functionality is provided by the distributed cloud component. The regional cloud platform requires the common components in Table 5-6, as well as the components in Table 5-7.

Components	Description
distributed-cloud	Provides centralized controller functions for management (e.g., lifecycle) of edge clouds.

Table 5-7. Regional cloud reference configuration profile.

5.2.3 Edge Cloud Reference Configuration Profile

Based on the hardware acceleration choice, the Linux kernel can be “standard” for inline GPU-based acceleration or “real-time” for look-aside FPGA-based acceleration. See Section 5.2.4 for the real-time Linux kernel configuration details. In addition, the edge cloud platform requires the common components in Table 5-6, as well as the components in Table 5-8.

Platform Components	Description
distributed-cloud-client	Provides edge cloud client services for distributed cloud function.
e1000e-kmod	E1000e driver for Intel® Gigabit NIC (optional – leveraged only if specific Intel hardware support is required).

gdrCOPY	Provides GPU memory copy library (optional – leveraged only if specific Nvidia hardware support is required).
i40e-kmod	I40e driver for Intel® 40G NIC (optional – leveraged only if specific Intel hardware support is required).
iavf-kmod	Driver for Intel® Ethernet adaptive virtual function (optional – leveraged only if specific Intel hardware support is required).
integrity-kmod	Kernel module for IMA
ixgbe-kmod	Ixgbe driver for Intel® 82598 and 82599 based 10G NIC (optional – leveraged only if specific Intel hardware support is required).
ixgbev-kmod	Driver for Intel® Ethernet adaptive virtual function for 10 Gig NIC (optional – leveraged only if specific Intel hardware support is required).
libibverbs-31mlnx1-ofed	Allows user-space processes to use RDMA verbs directly (optional – leveraged only if specific Mellanox hardware support is required).
mlnx-ofa-kernel	Driver for Mellanox NICs (optional – leveraged only if specific Mellanox hardware support is required).
opae-intel-fpga-driver-kmod	Open Programmable Acceleration Engine (OPAE) Kernel module for Intel FPGA.
qat17	Driver for Intel® QuickAssist Technology (optional – leveraged only if specific Intel hardware support is required).
rdma-core-45mlnx1	Driver for Mellanox RDMA-based interconnect (optional – leveraged only if specific Mellanox hardware support is required)
Linuxptp	Provides Precision Time Protocol according to IEEE standard 1588 and telecom profiles G.8265.1, G.8275.1 and G.8275.2 linuxptp.sourceforge.net
Kubernetes Plugin/Operators	Description
Multus	Multus is a Multi CNI plugin to support the Multi Networking feature in Kubernetes using CRD-based network objects in Kubernetes. The Multus plugin should be used on the master on all nodes to enable the use of multiple network interfaces inside a K8 pod. github.com/intel/multus-cni .
FPGA	FPGA Device plugin for FPGA passthrough (optional – leveraged only if specific Intel hardware support is required). github.com/intel/intel-device-plugins-for-kubernetes#fpga-device-plugin
QAT	QAT Device plugin for Intel® QAT adapters (optional – leveraged only if specific Intel hardware support is required). github.com/intel/intel-device-plugins-for-kubernetes/blob/master/cmd/qat_plugin/
SR-IOV	The SR-IOV plugin is required to enable the assignment of VFs to containers. SR-IOV provides a low-latency interface for both packet I/O and acceleration interfaces. github.com/hustcat/sriov-cni
CPU Manager	The CPU manager for Kubernetes can be used to provide basic core affinity for VNFs and CNFs. It is required to pin real-time threads to specific CPU cores in order to improve performance and meet real-time latency.

	github.com/intel/CPU-Manager-for-Kubernetes .
Node Feature Discovery	The Node Feature Discovery plugin enables discovering server hardware node features for Kubernetes. It detects hardware features available on each node in a Kubernetes cluster, and advertises those features using node labels. github.com/kubernetes-sigs/node-feature-discovery
Node Topology Manager	Node Topology Manager can be used to support NUMA awareness and pinning via the scheduler. kubernetes.io/docs/tasks/administer-cluster/topology-manager/
GPU Operator	The GPU operator manages NVIDIA GPU and automates tasks related to bootstrapping GPU enabled nodes (optional – leveraged only if specific Nvidia hardware support is required). nvidia.github.io/gpu-operator/

Table 5-8 Edge Cloud Reference Configuration Profile

5.2.4 Example Real-time Linux Configurations

Table 5-9 shows the reference configurations for the real-time Linux OS.

Linux Kernel	3.10.0-957.21.3.rt56.935.el7.x86_64	
Host or Guest Linux Kernel Parameters	rcu_nocbs=1-N rcu_nocb_poll isolcpus=1-N irqaffinity=0 tsc=perfect nohz_full=1-N idle=poll noswap	N is the total number of logical CPU cores. Use the “tuna” tool, provided in some Linux distributions, to configure run-time CPU isolation and thread priorities, if permanent isolation with isolcpus kernel option is not desired.
	selinux=0 enforcing=0	Disable SELinux policies can help improve latency but should only be done if the security considerations are fully understood.
	default_hugepagesz=1G hugepagesz=1G hugepages=21	Use huge pages to minimize TLB miss overhead; use 2MB huge pages when system memory is limited.
	usbcore.autosuspend=-1 nmi_watchdog=0 softlockup_panic=0 cgroup_disable=memory skew_tick=1 nosoftlockup	General configurations for real time.
	kthread_cpus=0	Limit host kernel threads to the OS core. Used also in guest assuming a pure user space application design.
	intel_pstate=disable intel_idle.max_cstate=0 processor.max_cstate=1	Disabling P-states and limiting C-states in the kernel helps improve real time performance.
Host Linux GRUB Parameters	intel_iommu=on iommu=pt pci=pcie_bus_perf	Enable SR-IOV and optimal PCI performance.
Host and Guest Tuning	Turn off graphical desktop and unnecessary daemons in the system	
	If ext3 or ext4 file system is used, consider using noatime in fstab	

Host Kernel Command Line example	<pre> BOOT_IMAGE=/vmlinuz-3.10.0-957.21.3.rt56.935.el7.x86_64 root=/dev/mapper/centos- root ro crashkernel=auto nomodeset rd.lvm.lv=centos/root rd.lvm.lv=centos/swap rhgb quiet hugepagesz=1G hugepages=35 default_hugepagesz=1G intel_iommu=on iommu=pt LANG=en_GB.UTF-8 tboot=false biosdevname=0 usbcore.autosuspend=-1 selinux=0 enforcing=0 nmi_watchdog=0 softlockup_panic=0 cgroup_disable=memory skew_tick=1 isolcpus=1-N rcu_nocbs=1-N kthread_cpus=0 irqaffinity=0 nohz_full=1-N nopti clock=pit no_timer_check pci=pcie_bus_perf clocksource=tsc tsc=perfect idle=poll intel_pstate=disable intel_idle.max_cstate=0 processor.max_cstate=1 nohz=on mce=off nosoftlockup </pre>
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Table 5-9. Real-time Linux configurations.

5.3 Time Synchronization Reference Design

Figure 5-7 illustrates an example of the redundant synchronization network for an edge site aligned with the requirements. The O-RUs are expected to be connected to the same leaf-spine fabric, and as such the O-Cloud network is considered to also be part of the fronthaul network.

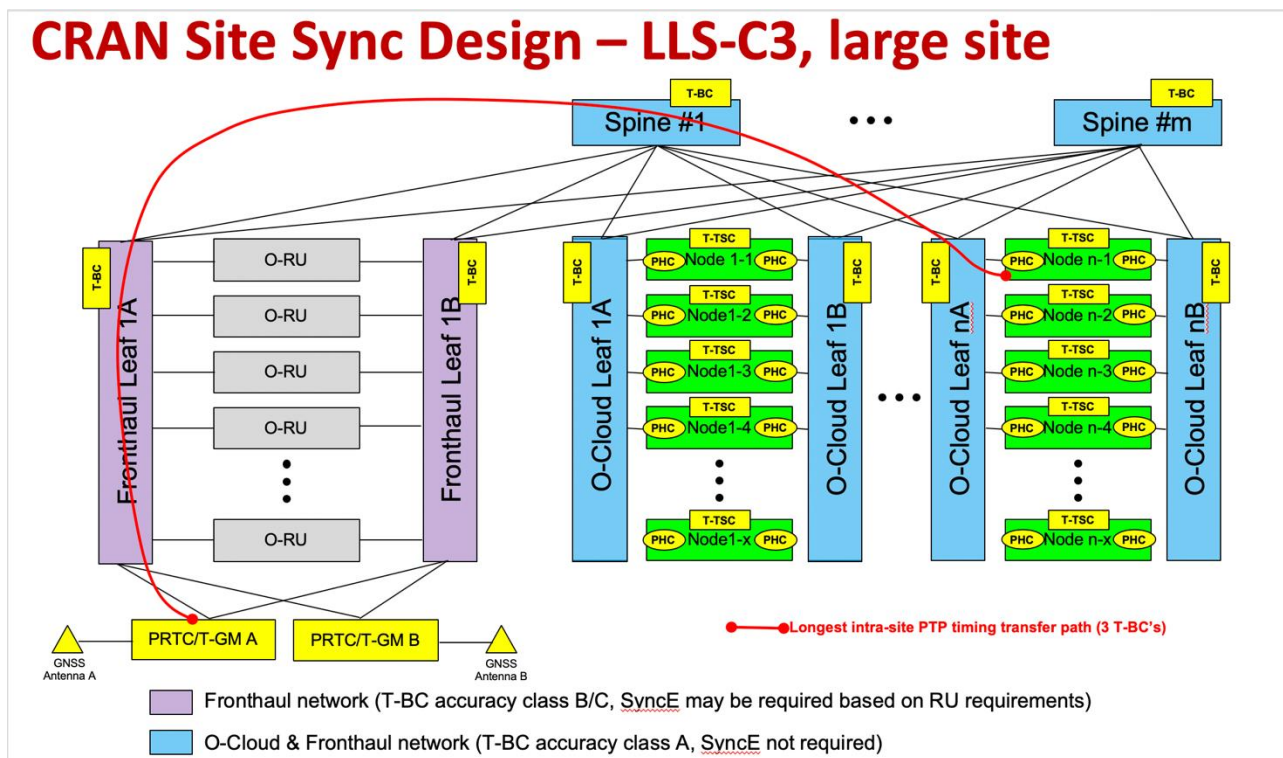


Figure 5-7. Key synchronization chain elements in an example implementation in O-Cloud for LLS-C3.

The salient features of this design example are:

- The synchronization reference sources (PRTC/T-GM) are redundant and located in the edge cloud. In this example, the references are derived from the GNSS system, but any other implementation compliant with the CUS specification [2] requirements (and references therein), which acts as G.8275.1 T-GM, is acceptable.
- The synchronization reference sources are connected to the leaf-spine fabric with redundant Ethernet interconnects. Specifically, each grandmaster connects to at least two leaf switches.
- For larger site configurations (i.e. in which the switch capacity and port count requirements exceed the capabilities of the single physical switch), the structure is expanded in a “leaf-spine” fabric configuration, as illustrated in [2]. For most edge site configurations, it is expected that a two-tier structure is sufficient, which implies that the maximum length of any active synchronization transfer path from co-located sources is three boundary clocks between the grandmaster and any node in the edge cloud. Therefore, the synchronization path “length” in terms of number of boundary clocks that needs to be supportable in this configuration is typically from one to three. Capability to support more nodes in the path is desirable to allow for configurations where the grandmasters are not co-located at the edge cloud.

- All edge O-Cloud nodes are attached to at least two physically separated, independent leaf switches to provide network redundancy and tolerance against a single point of failure that can affect more than one node (i.e. switch failures). In addition to normal network fault tolerance, this is also useful for maintaining the synchronization network fault tolerance.

Many alternative variations and physical realizations of the above design are possible (e.g. PRTCs/GMs can be combined with switches or routers, leaf-spine structure can be integrated onto physical box, sizes of elements can vary, etc.), but functionally with respect to the synchronization transfer chain, all of the entities are expected to be present (i.e. PRTC/T-GMs, Telecom Boundary Clock nodes, as well Telecom subordinate Clock nodes).

For this design example, the O-DU synchronization performance requirements as specified in the CUS specification [2] for the O-DU LLS-C3 configuration expected to be met with the following set of elements in the synchronization transfer path:

- T-GM A and T-GM B clocks with performance class ITU-T G.8272 PRTC/T-GM type PRTC-B serving as grandmasters for O-DUs as well as O-RUs (O-DUs require PRTC-A or better while O-RUs require PRTC-B)
- The Fronthaul Leafs acting as T-BCs in both the O-RU and O-DU synchronization path, with support for a minimum performance class ITU-T G.8273.2 T-BC Class B (Class C is recommended);
- The Spine and O-Cloud Leaf switches acting as T-BCs in the O-DU synchronization path only, with the least demanding T-BC performance class ITU-T G.8273.2 T-BC Class A or better;
- T-TSC implementation on the O-Cloud nodes with the least demanding performance class ITU-T G.8273.2, Annex C, T-TSC Class A or better.

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1.2 “Compliant Implementation” means any system, device, method or operation (whether implemented in hardware, software or combinations thereof) that fully conforms to a Final Specification.

1.3 “Adopter(s)” means all entities, who are not Members, Contributors or Academic Contributors, including their Affiliates, who wish to download, use or otherwise access O-RAN Specifications.

1.4 “Minor Update” means an update or revision to an O-RAN Specification published by O-RAN ALLIANCE that does not add any significant new features or functionality and remains interoperable with the prior version of an O-RAN Specification. The term “O-RAN Specifications” includes Minor Updates.

1.5 “Necessary Claims” means those claims of all present and future patents and patent applications, other than design patents and design registrations, throughout the world, which (i) are owned or otherwise licensable by a Member, Contributor or Academic Contributor during the term of its Member, Contributor or Academic Contributorship; (ii) such Member, Contributor or Academic Contributor has the right to grant a license without the payment of consideration to a third party; and (iii) are necessarily infringed by a Compliant Implementation (without considering any Contributions not included in the Final Specification). A claim is necessarily infringed only when it is not possible on technical (but not commercial) grounds, taking into account normal technical practice and the state of the art generally available at the date any Final Specification was published by the O-RAN ALLIANCE or the date the patent claim first came into existence, whichever last occurred, to make, sell, lease, otherwise dispose of, repair, use or operate a Compliant Implementation without infringing that claim. For the avoidance of doubt in exceptional cases where a Final Specification can only be implemented by technical solutions, all of which infringe patent claims, all such patent claims shall be considered Necessary Claims.

1.6 “Defensive Suspension” means for the purposes of any license grant pursuant to Section 3, Member, Contributor, Academic Contributor, Adopter, or any of their Affiliates, may have the discretion to include in their license a term allowing the licensor to suspend the license against a licensee who brings a patent infringement suit against the licensing Member, Contributor, Academic Contributor, Adopter, or any of their Affiliates.

Section 2: COPYRIGHT LICENSE

2.1 Subject to the terms and conditions of this Agreement, O-RAN ALLIANCE hereby grants to Adopter a nonexclusive, nontransferable, irrevocable, non-sublicensable, worldwide copyright license to obtain, use and modify O-RAN Specifications, but not to further distribute such O-RAN Specification in any modified or unmodified way, solely in furtherance of implementations of an ORAN

Specification.

2.2 Adopter shall not use O-RAN Specifications except as expressly set forth in this Agreement or in a separate written agreement with O-RAN ALLIANCE.

Section 3: FRAND LICENSE

3.1 Members, Contributors and Academic Contributors and their Affiliates are prepared to grant based on a separate Patent License Agreement to each Adopter under Fair Reasonable And Non- Discriminatory (FRAND) terms and conditions with or without compensation (royalties) a nonexclusive, non-transferable, irrevocable (but subject to Defensive Suspension), non-sublicensable, worldwide patent license under their Necessary Claims to make, have made, use, import, offer to sell, lease, sell and otherwise distribute Compliant Implementations; provided, however, that such

license shall not extend: (a) to any part or function of a product in which a Compliant Implementation is incorporated that is not itself part of the Compliant Implementation; or (b) to any Adopter if that Adopter is not making a reciprocal grant to Members, Contributors and Academic Contributors, as set forth in Section 3.3. For the avoidance of doubt, the foregoing licensing commitment includes the distribution by the Adopter's distributors and the use by the Adopter's customers of such licensed Compliant Implementations.

3.2 Notwithstanding the above, if any Member, Contributor or Academic Contributor, Adopter or their Affiliates has reserved the right to charge a FRAND royalty or other fee for its license of Necessary Claims to Adopter, then Adopter is entitled to charge a FRAND royalty or other fee to such Member, Contributor or Academic Contributor, Adopter and its Affiliates for its license of Necessary Claims to its licensees.

3.3 Adopter, on behalf of itself and its Affiliates, shall be prepared to grant based on a separate Patent License Agreement to each Members, Contributors, Academic Contributors, Adopters and their Affiliates under Fair Reasonable And Non-Discriminatory (FRAND) terms and conditions with or without compensation (royalties) a nonexclusive, non-transferable, irrevocable (but subject to Defensive Suspension), non-sublicensable, worldwide patent license under their Necessary Claims to make, have made, use, import, offer to sell, lease, sell and otherwise distribute Compliant Implementations; provided, however, that such license will not extend: (a) to any part or function of a product in which a Compliant Implementation is incorporated that is not itself part of the Compliant Implementation; or (b) to any Members, Contributors, Academic Contributors, Adopters and their Affiliates that is not making a reciprocal grant to Adopter, as set forth in Section 3.1. For the avoidance of doubt, the foregoing licensing commitment includes the distribution by the Members', Contributors', Academic Contributors', Adopters' and their Affiliates' distributors and the use by the Members', Contributors', Academic Contributors', Adopters' and their Affiliates' customers of such licensed Compliant Implementations.

Section 4: TERM AND TERMINATION

4.1 This Agreement shall remain in force, unless early terminated according to this Section 4.

4.2 O-RAN ALLIANCE on behalf of its Members, Contributors and Academic Contributors may terminate this Agreement if Adopter materially breaches this Agreement and does not cure or is not capable of curing such breach within thirty (30) days after being given notice specifying the breach.

4.3 Sections 1, 3, 5 - 11 of this Agreement shall survive any termination of this Agreement. Under surviving Section 3, after termination of this Agreement, Adopter will continue to grant licenses (a) to entities who become Adopters after the date of termination; and (b) for future versions of ORAN Specifications that are backwards compatible with the version that was current as of the date of termination.

Section 5: CONFIDENTIALITY

Adopter will use the same care and discretion to avoid disclosure, publication, and dissemination of O-RAN Specifications to third parties, as Adopter employs with its own confidential information, but no less than reasonable care. Any disclosure by Adopter to its Affiliates, contractors and consultants should be subject to an obligation of confidentiality at least as restrictive as those contained in this Section. The foregoing obligation shall not apply to any information which is: (1) rightfully known by Adopter without any limitation on use or disclosure prior to disclosure; (2) publicly available through no fault of Adopter; (3) rightfully received without a duty of confidentiality; (4) disclosed by O-RAN ALLIANCE or a Member, Contributor or Academic Contributor to a third party without a duty of confidentiality on such third party; (5) independently developed by Adopter; (6) disclosed pursuant to the order of a court or other authorized governmental body, or as required by law, provided that Adopter provides reasonable prior written notice to O-RAN ALLIANCE, and cooperates with O-RAN ALLIANCE and/or the applicable Member, Contributor or Academic Contributor to have the opportunity to oppose any such order; or (7) disclosed by Adopter with O-RAN ALLIANCE's prior written approval.

Section 6: INDEMNIFICATION

Adopter shall indemnify, defend, and hold harmless the O-RAN ALLIANCE, its Members, Contributors or Academic Contributors, and their employees, and agents and their respective successors, heirs and assigns (the "Indemnitees"), against any liability, damage, loss, or expense (including reasonable attorneys' fees and expenses) incurred by or imposed upon any of the Indemnitees in connection with any claims, suits, investigations, actions, demands or judgments arising out of Adopter's use of the licensed O-RAN Specifications or Adopter's commercialization of products that comply with O-RAN Specifications.

Section 7: LIMITATIONS ON LIABILITY; NO WARRANTY

EXCEPT FOR BREACH OF CONFIDENTIALITY, ADOPTER'S BREACH OF SECTION 3, AND ADOPTER'S INDEMNIFICATION OBLIGATIONS, IN NO EVENT SHALL ANY PARTY BE LIABLE TO ANY OTHER PARTY OR THIRD PARTY FOR ANY INDIRECT, SPECIAL, INCIDENTAL, PUNITIVE OR CONSEQUENTIAL DAMAGES RESULTING FROM ITS PERFORMANCE OR NON-PERFORMANCE UNDER THIS AGREEMENT, IN EACH CASE WHETHER UNDER CONTRACT, TORT, WARRANTY, OR OTHERWISE, AND WHETHER OR NOT SUCH PARTY HAD ADVANCE NOTICE OF THE POSSIBILITY OF SUCH DAMAGES. O-RAN SPECIFICATIONS ARE PROVIDED "AS IS" WITH NO WARRANTIES OR CONDITIONS WHATSOEVER, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE. THE O-RAN ALLIANCE AND THE MEMBERS, CONTRIBUTORS OR ACADEMIC CONTRIBUTORS EXPRESSLY DISCLAIM ANY WARRANTY OR CONDITION OF MERCHANTABILITY, SECURITY, SATISFACTORY QUALITY, NONINFRINGEMENT, FITNESS FOR ANY PARTICULAR PURPOSE, ERROR-FREE OPERATION, OR ANY WARRANTY OR CONDITION FOR O-RAN SPECIFICATIONS.

Section 8: ASSIGNMENT

Adopter may not assign the Agreement or any of its rights or obligations under this Agreement or make any grants or other sublicenses to this Agreement, except as expressly authorized hereunder, without having first received the prior, written consent of the O-RAN ALLIANCE, which consent may be withheld in O-RAN ALLIANCE's sole discretion. O-RAN ALLIANCE may freely assign this Agreement.

Section 9: THIRD-PARTY BENEFICIARY RIGHTS

Adopter acknowledges and agrees that Members, Contributors and Academic Contributors (including future Members, Contributors and Academic Contributors) are entitled to rights as a third-party beneficiary under this Agreement, including as licensees under Section 3.

Section 10: BINDING ON AFFILIATES

Execution of this Agreement by Adopter in its capacity as a legal entity or association constitutes that legal entity's or association's agreement that its Affiliates are likewise bound to the obligations that are applicable to Adopter hereunder and are also entitled to the benefits of the rights of Adopter hereunder.

Section 11: GENERAL

This Agreement is governed by the laws of Germany without regard to its conflict or choice of law provisions.

This Agreement constitutes the entire agreement between the parties as to its express subject matter and expressly supersedes and replaces any prior or contemporaneous agreements between the parties, whether written or oral, relating to the subject matter of this Agreement.

Adopter, on behalf of itself and its Affiliates, agrees to comply at all times with all applicable laws, rules and regulations with respect to its and its Affiliates' performance under this Agreement, including without limitation, export control and antitrust laws. Without limiting the generality of the foregoing, Adopter acknowledges that this Agreement prohibits any communication that would violate the antitrust laws.

By execution hereof, no form of any partnership, joint venture or other special relationship is created between Adopter, or O-RAN ALLIANCE or its Members, Contributors or Academic Contributors. Except as expressly set forth in this Agreement, no party is authorized to make any commitment on behalf of Adopter, or O-RAN ALLIANCE or its Members, Contributors or Academic Contributors.

In the event that any provision of this Agreement conflicts with governing law or if any provision is held to be null, void or otherwise ineffective or invalid by a court of competent jurisdiction, (i) such provisions will be deemed stricken from the contract, and (ii) the remaining terms, provisions, covenants and restrictions of this Agreement will remain in full force and effect.

Any failure by a party or third party beneficiary to insist upon or enforce performance by another party of any of the provisions of this Agreement or to exercise any rights or remedies under this Agreement or otherwise by law shall not be construed as a waiver or relinquishment to any extent of the other parties' or third party beneficiary's right to assert or rely upon any such provision, right or remedy in that or any other instance; rather the same shall be and remain in full force and effect.